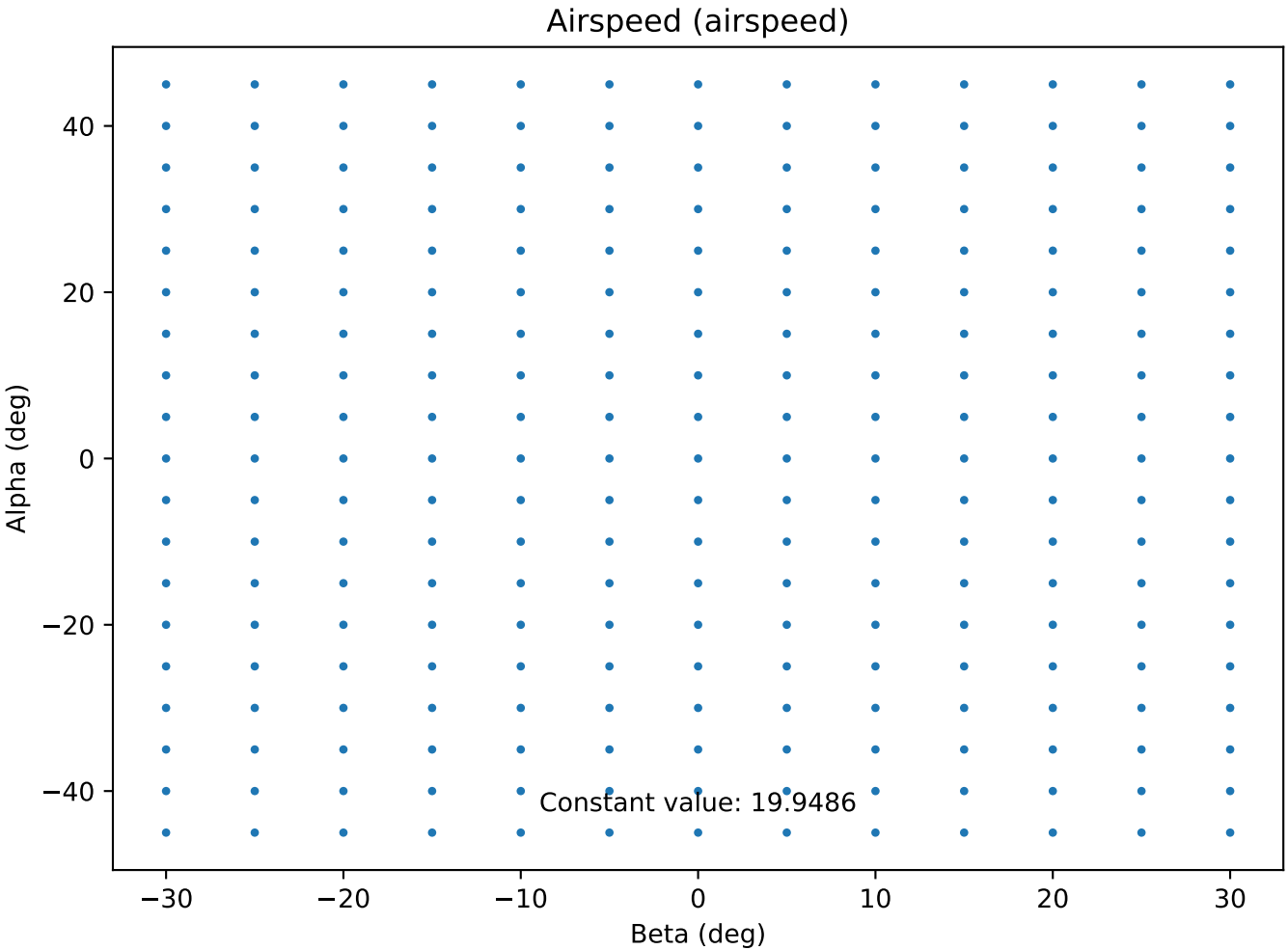
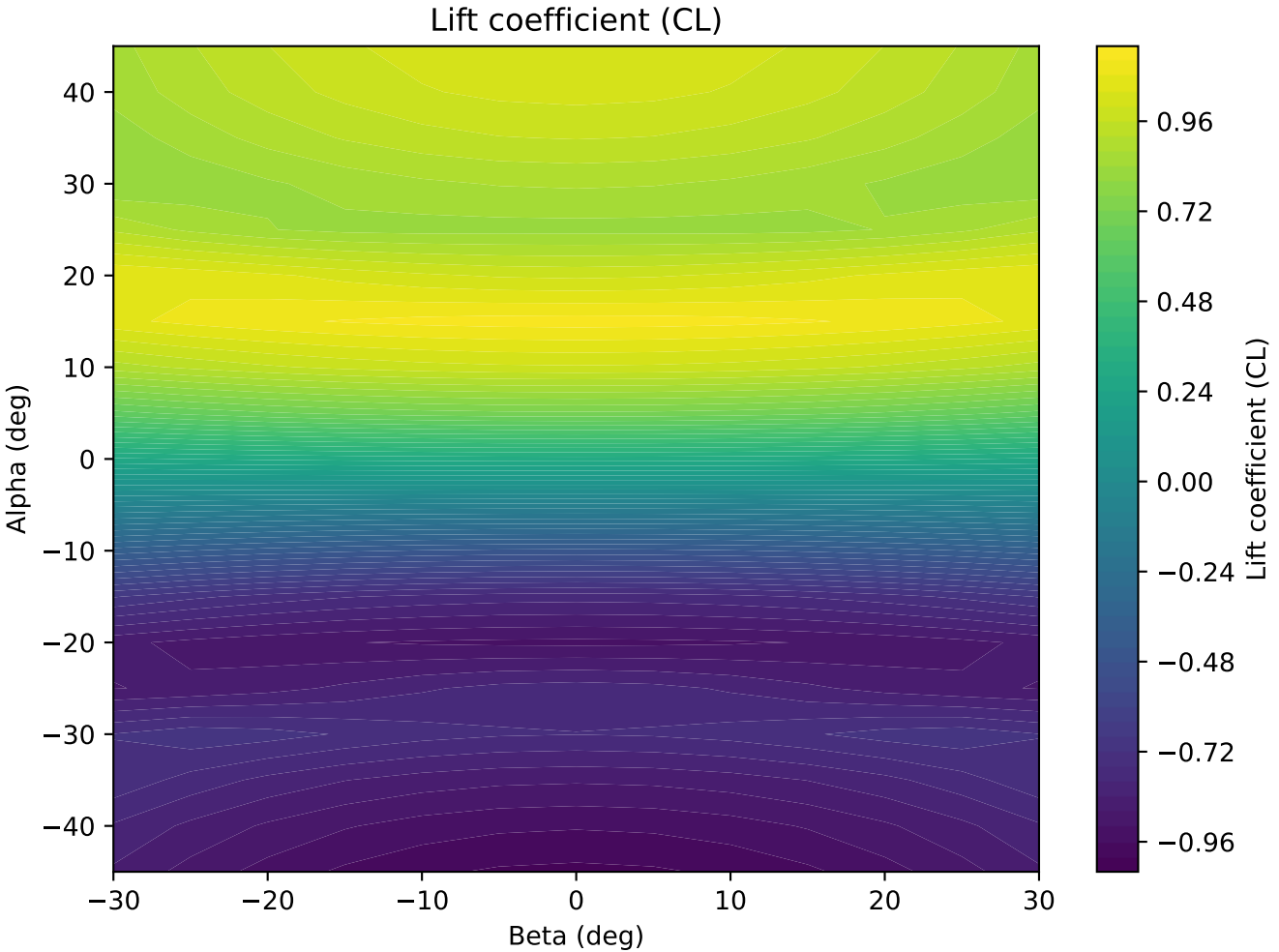
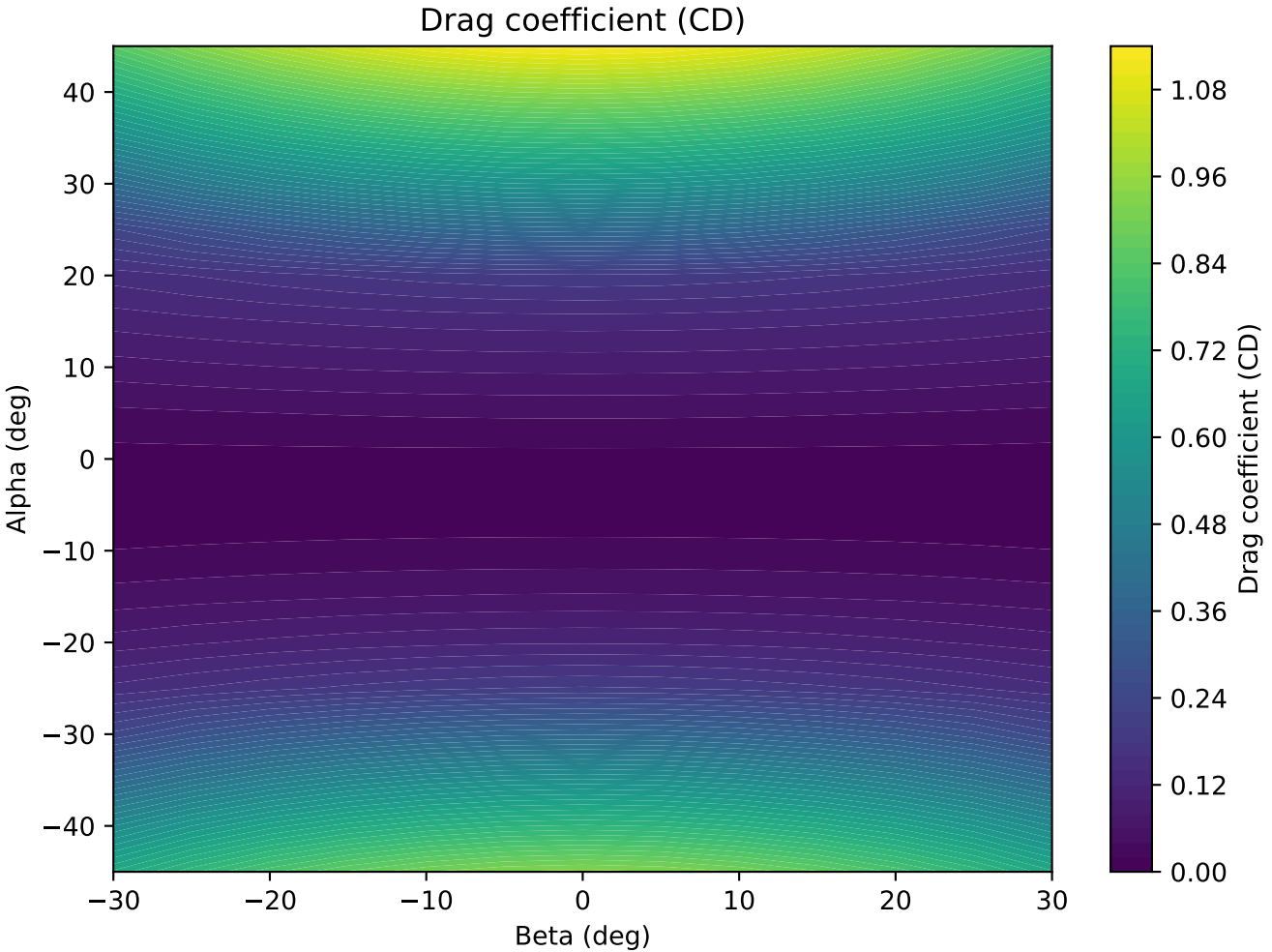
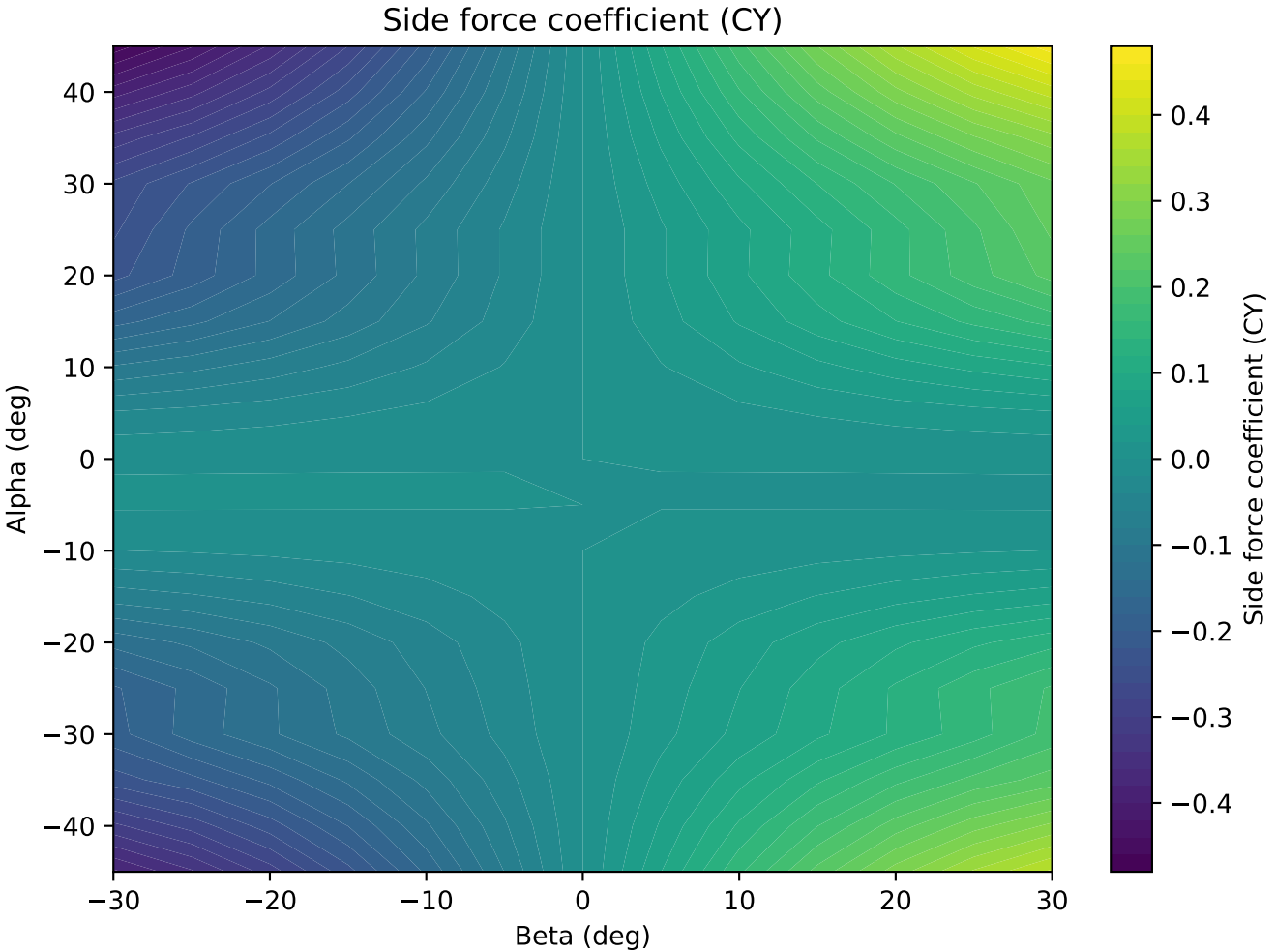
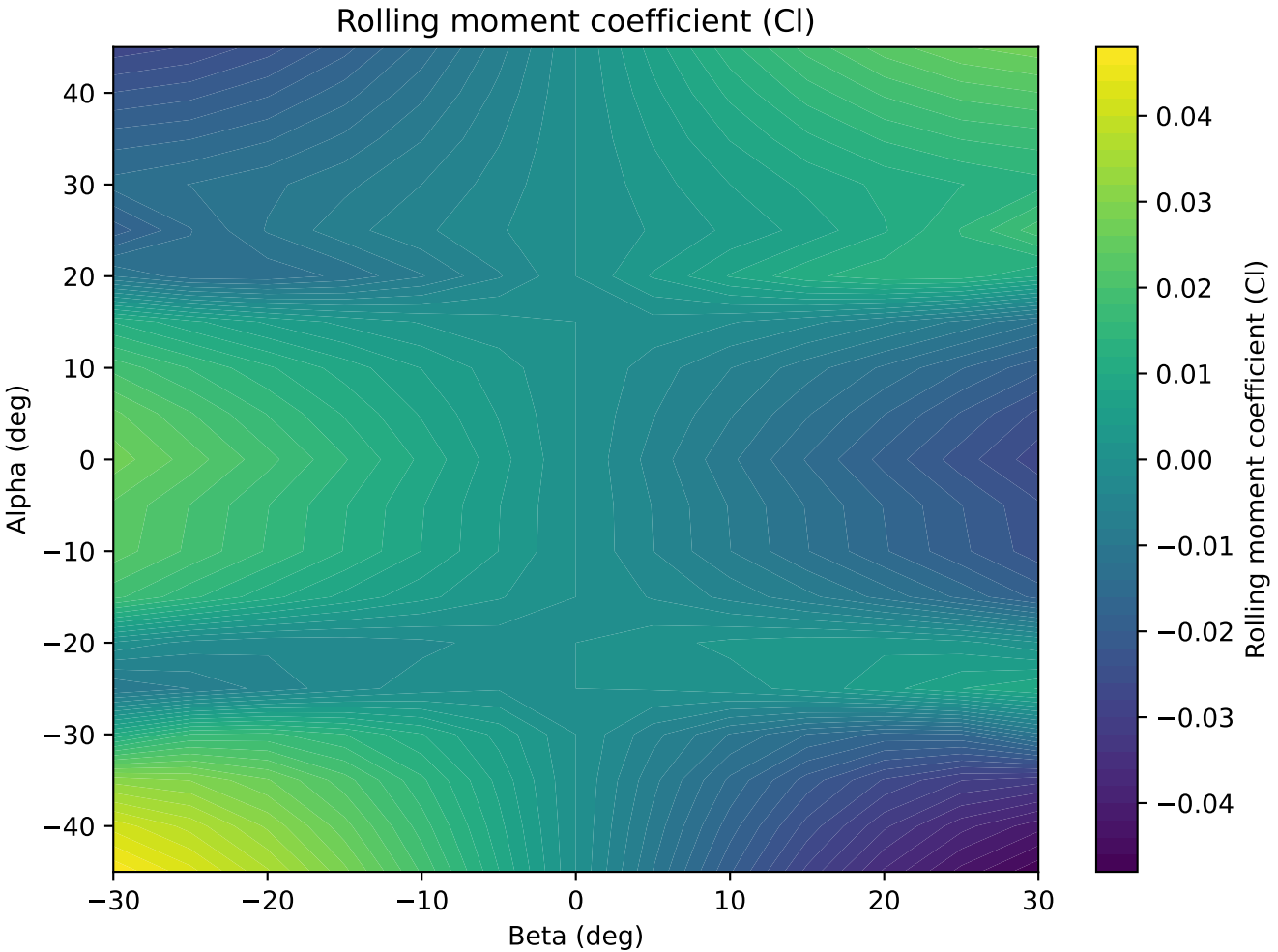
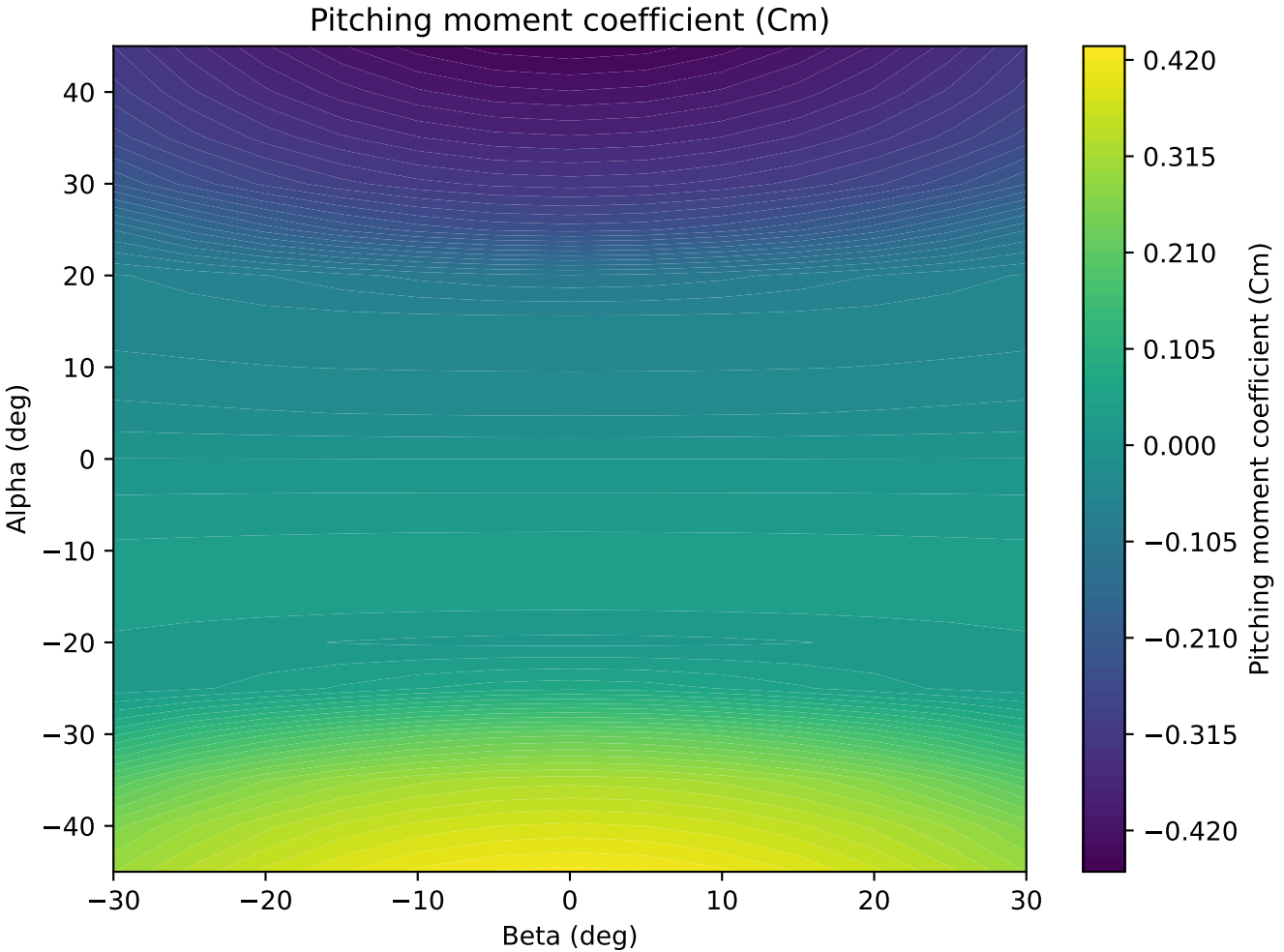

Airspeed (airspeed)
Alpha (deg)
Beta (deg)
40
20
0
−20
−40
−30
−20
−10
0
10
20
30
Constant value: 19.9486

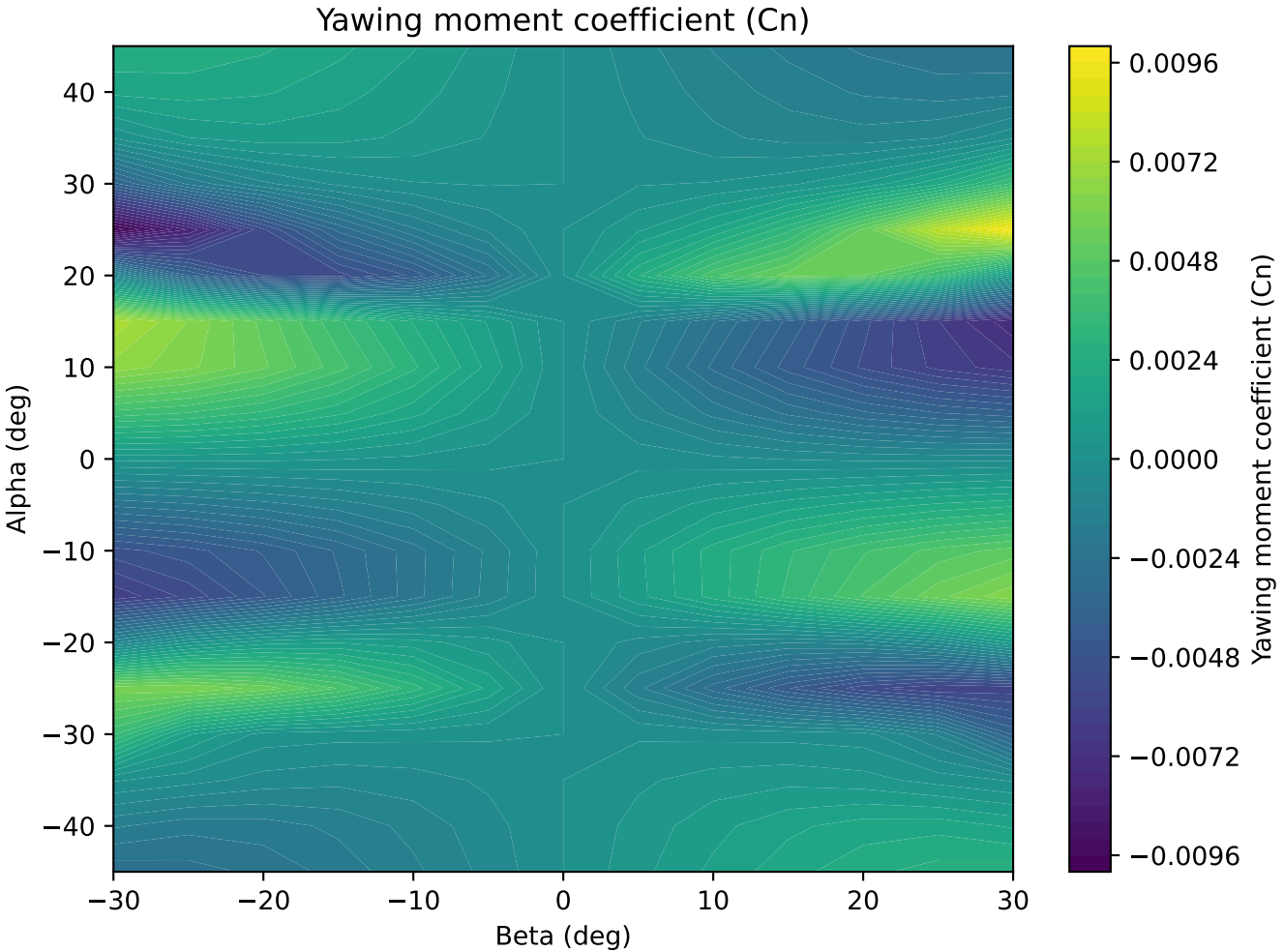

Lift coefficient (CL)
Alpha (deg)
Beta (deg)
Lift coefficient (CL)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.96
0.72
0.48
0.24
0.00
−0.24
−0.48
−0.72
−0.96

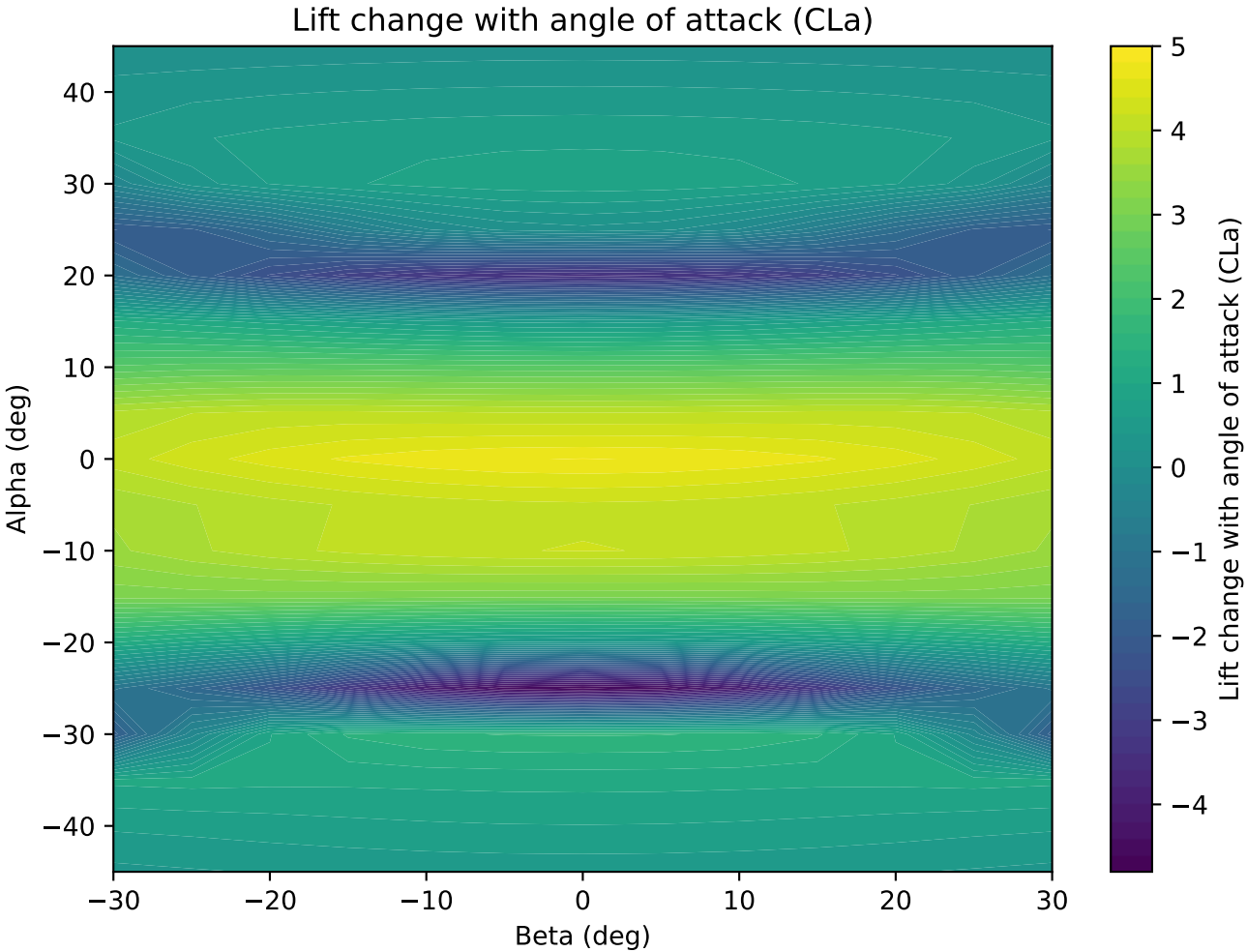

Drag coefficient (CD)
Alpha (deg)
Beta (deg)
Drag coefficient (CD)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
1.08
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

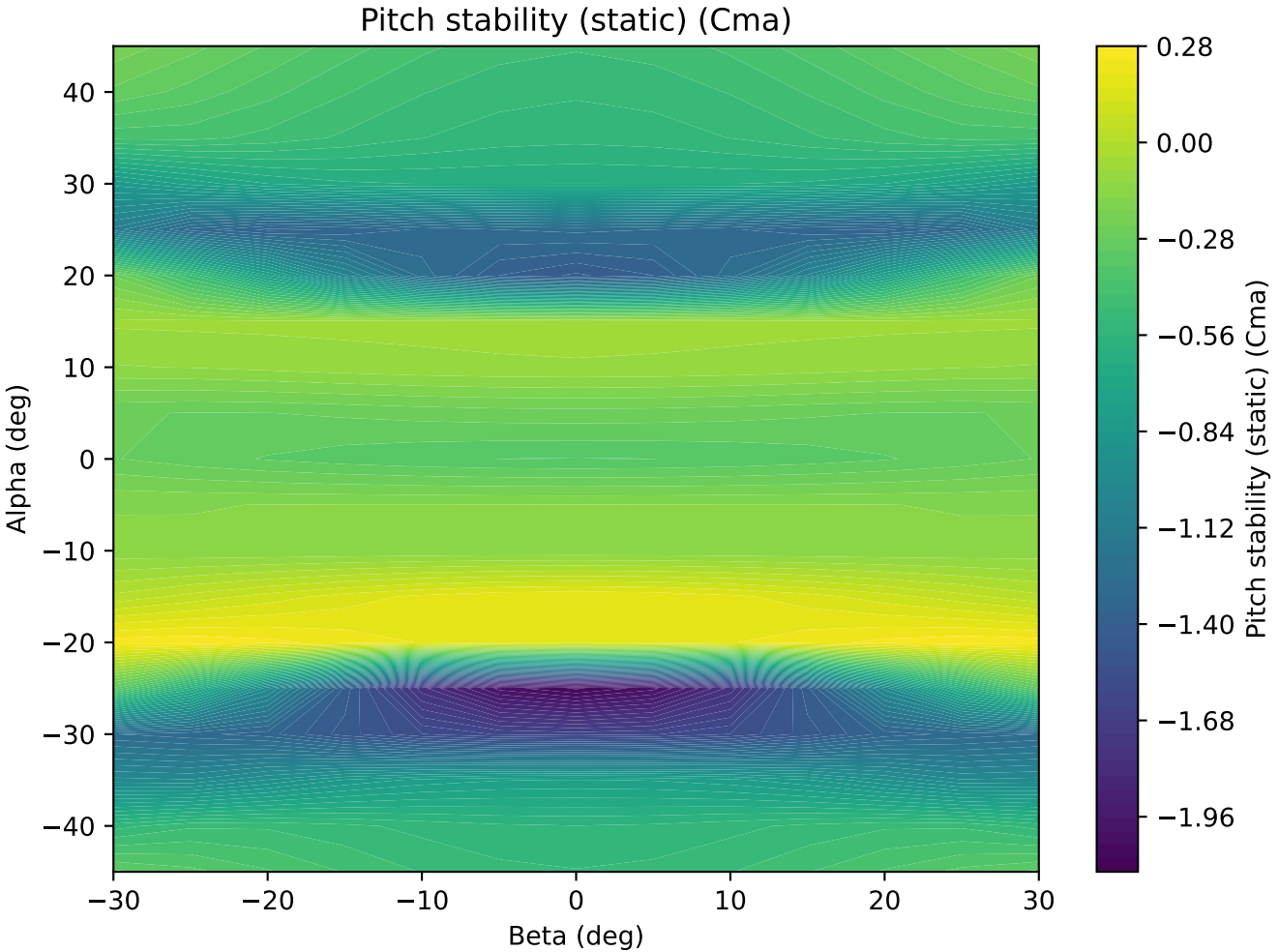

Side force coefficient (CY)
Alpha (deg)
Beta (deg)
Side force coefficient (CY)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4

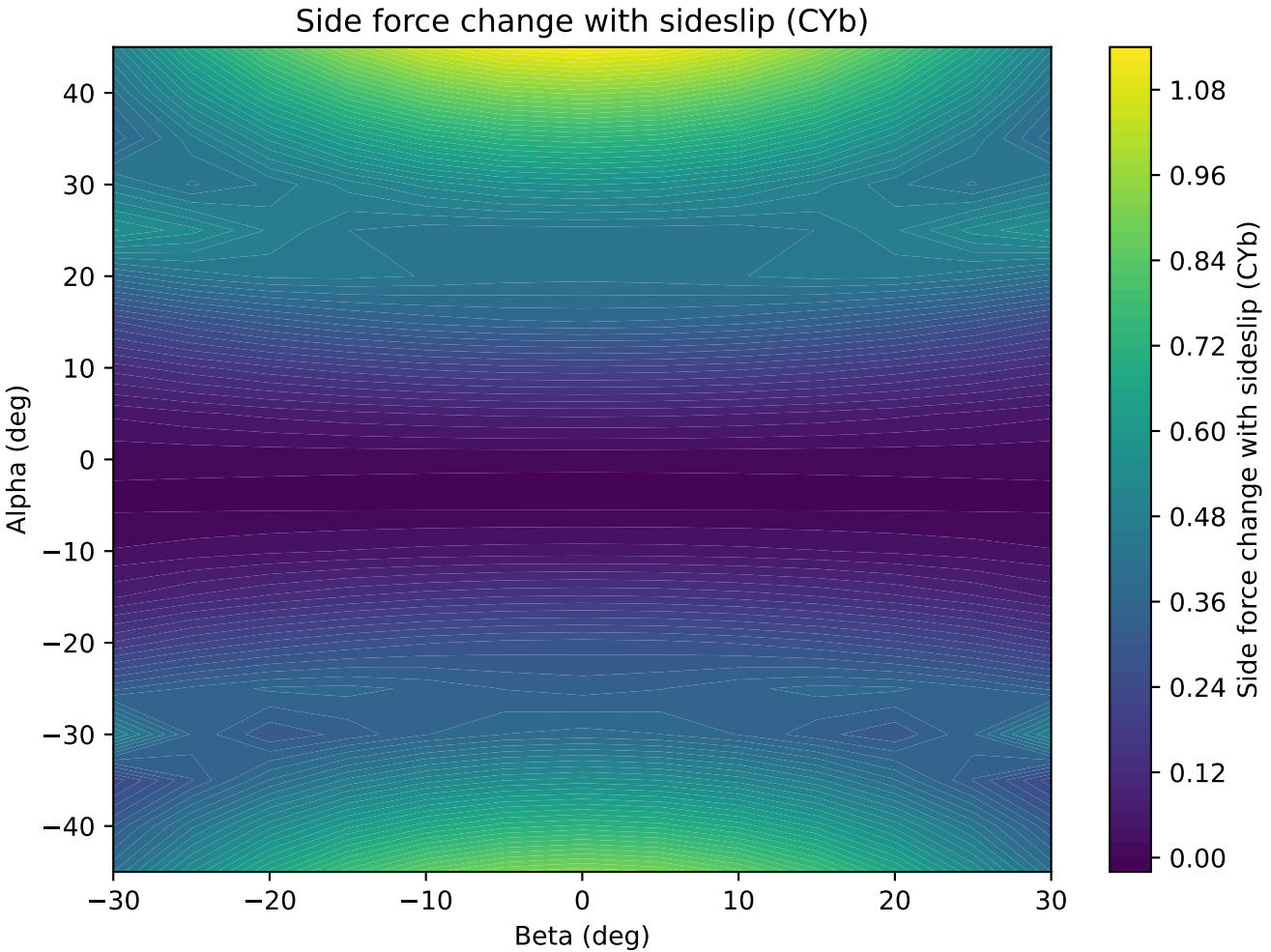

Rolling moment coefficient (Cl)
Alpha (deg)
Beta (deg)
Rolling moment coefficient (Cl)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04

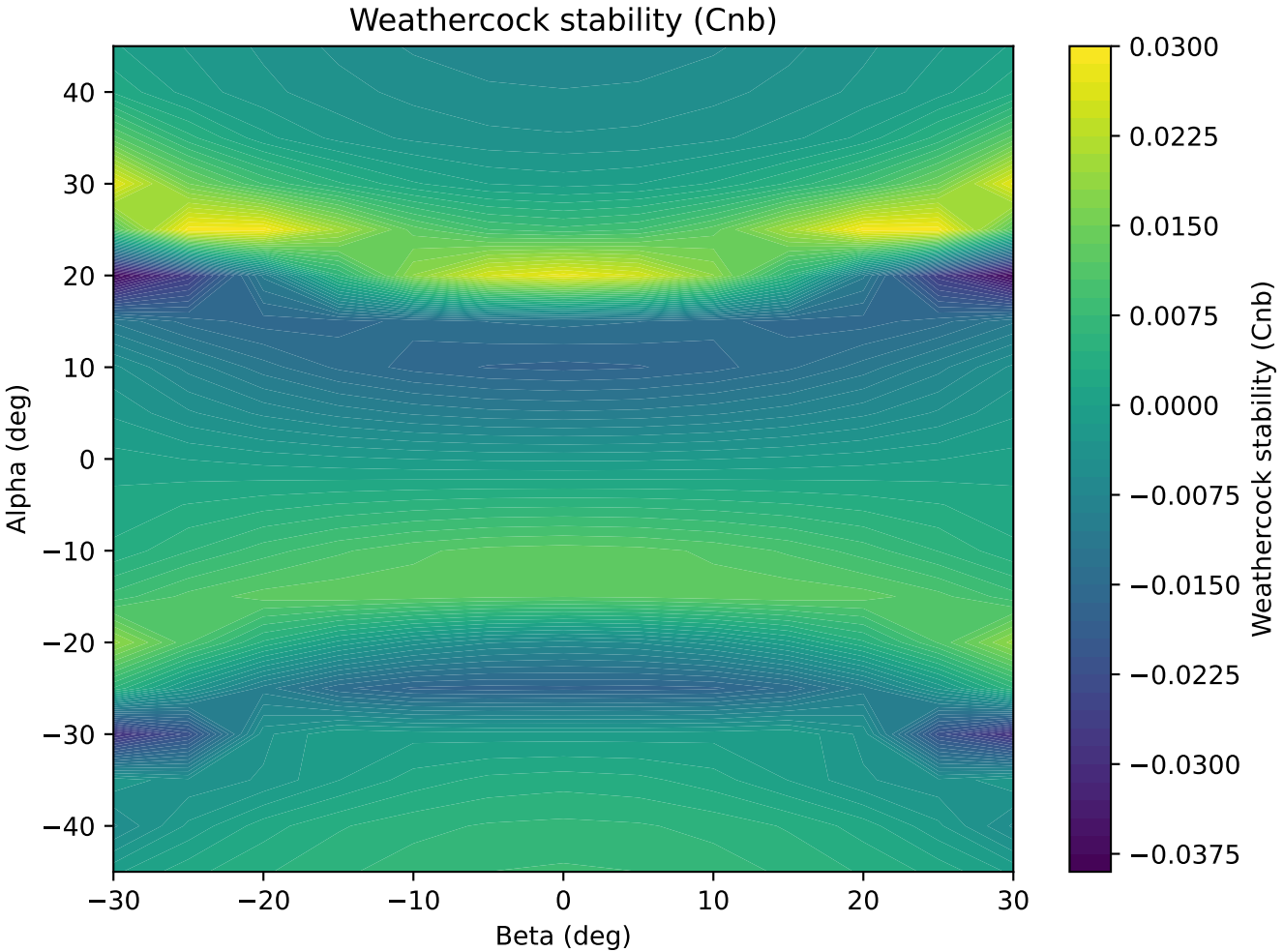

Pitching moment coefficient (Cm)
Alpha (deg)
Beta (deg)
Pitching moment coefficient (Cm)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.420
0.315
0.210
0.105
0.000
−0.105
−0.210
−0.315
−0.420

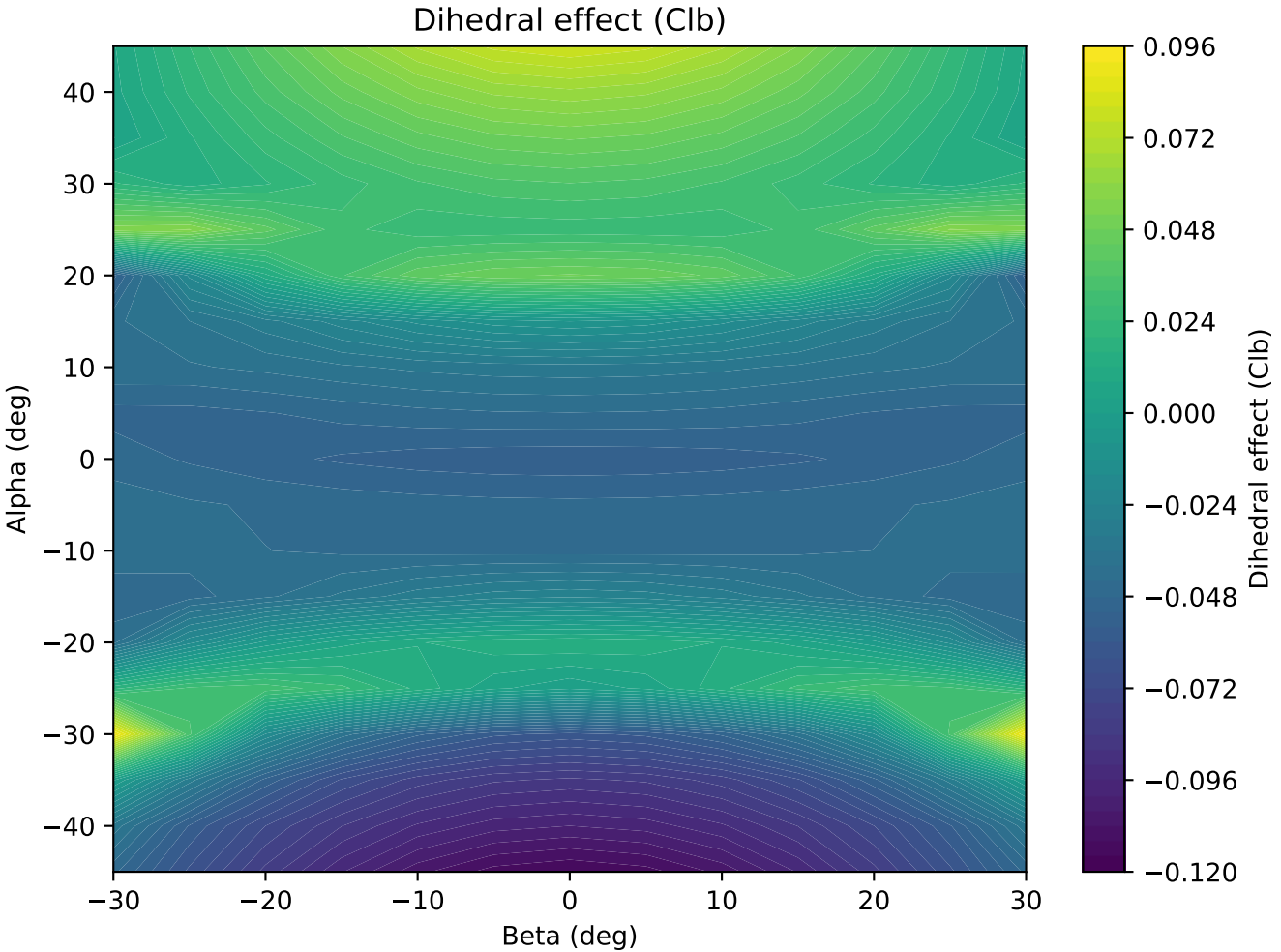

Yawing moment coefficient (Cn)
Alpha (deg)
Beta (deg)
Yawing moment coefficient (Cn)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.0096
0.0072
0.0048
0.0024
0.0000
−0.0024
−0.0048
−0.0072
−0.0096

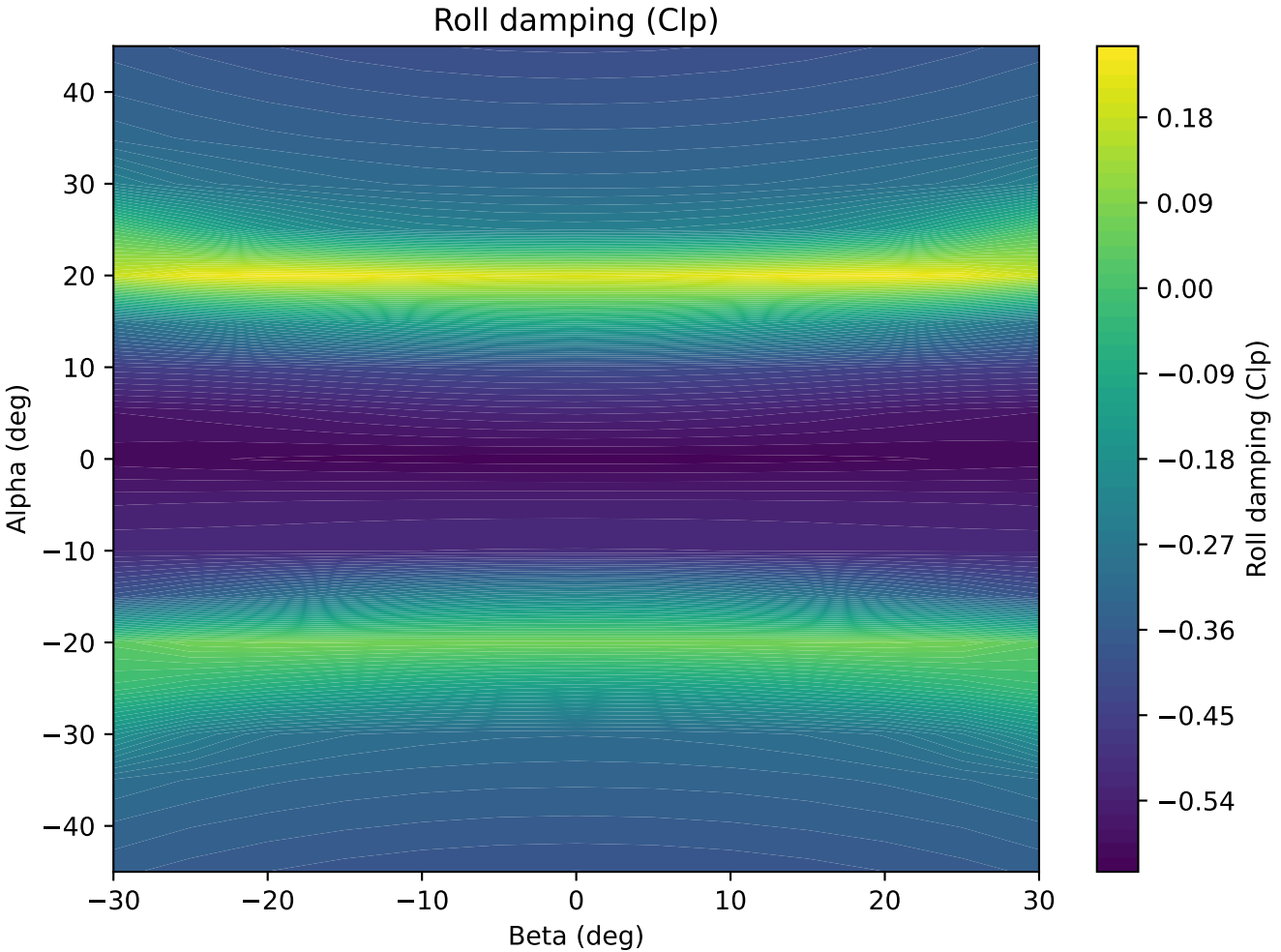

Lift change with angle of attack (CLa)
Alpha (deg)
Beta (deg)
Lift change with angle of attack (CLa)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
5
4
3
2
1
0
−1
−2
−3
−4

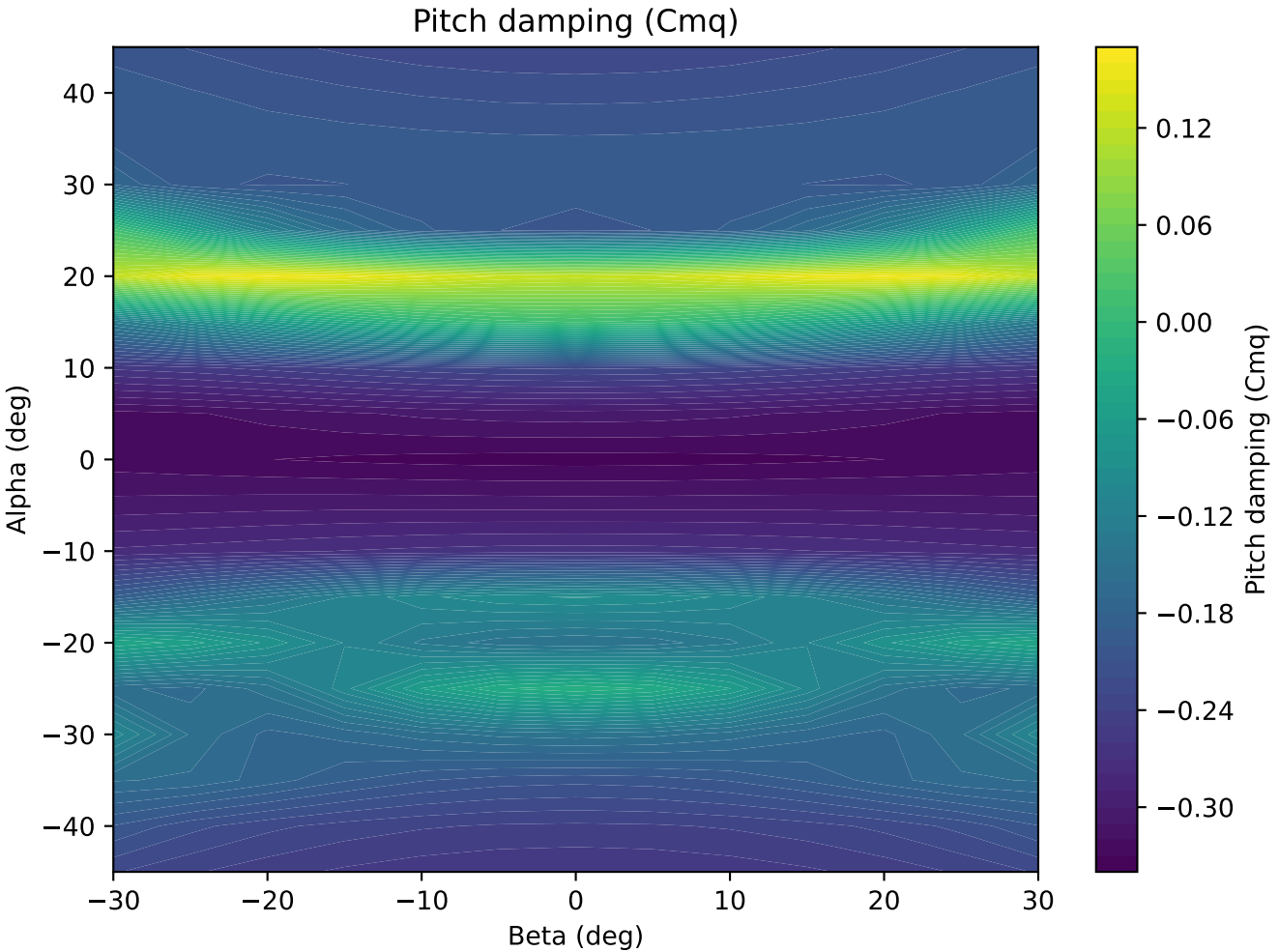

Pitch stability (static) (Cma)
Alpha (deg)
Beta (deg)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.28
0.00
−0.28
−0.56
−0.84
−1.12
−1.40
−1.68
−1.96
Pitch stability (static) (Cma)

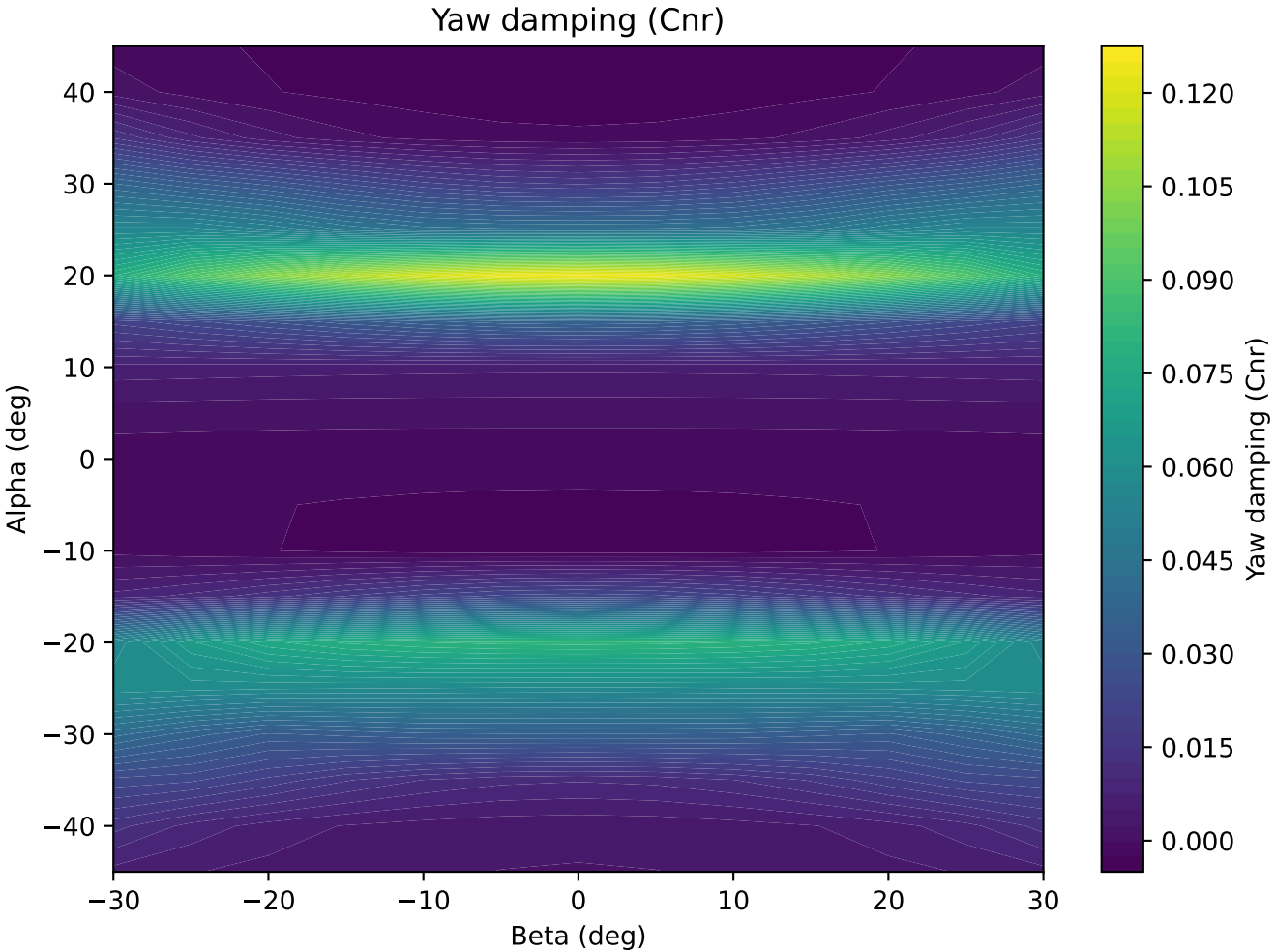

Side force change with sideslip (CYb)
Alpha (deg)
Beta (deg)
Side force change with sideslip (CYb)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
1.08
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

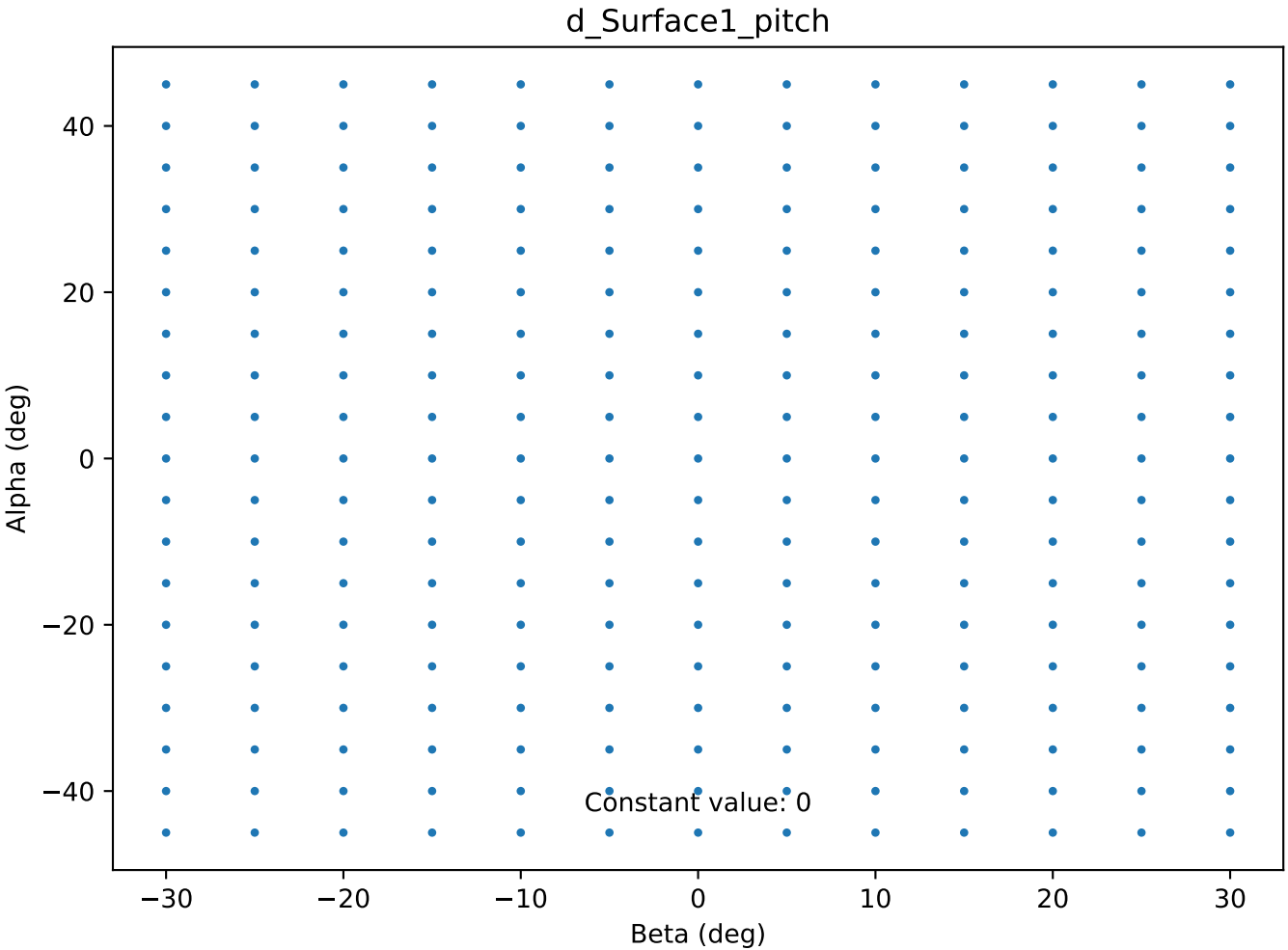

Weathercock stability (Cnb)
Alpha (deg)
Beta (deg)
Weathercock stability (Cnb)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.0300
0.0225
0.0150
0.0075
0.0000
−0.0075
−0.0150
−0.0225
−0.0300
−0.0375

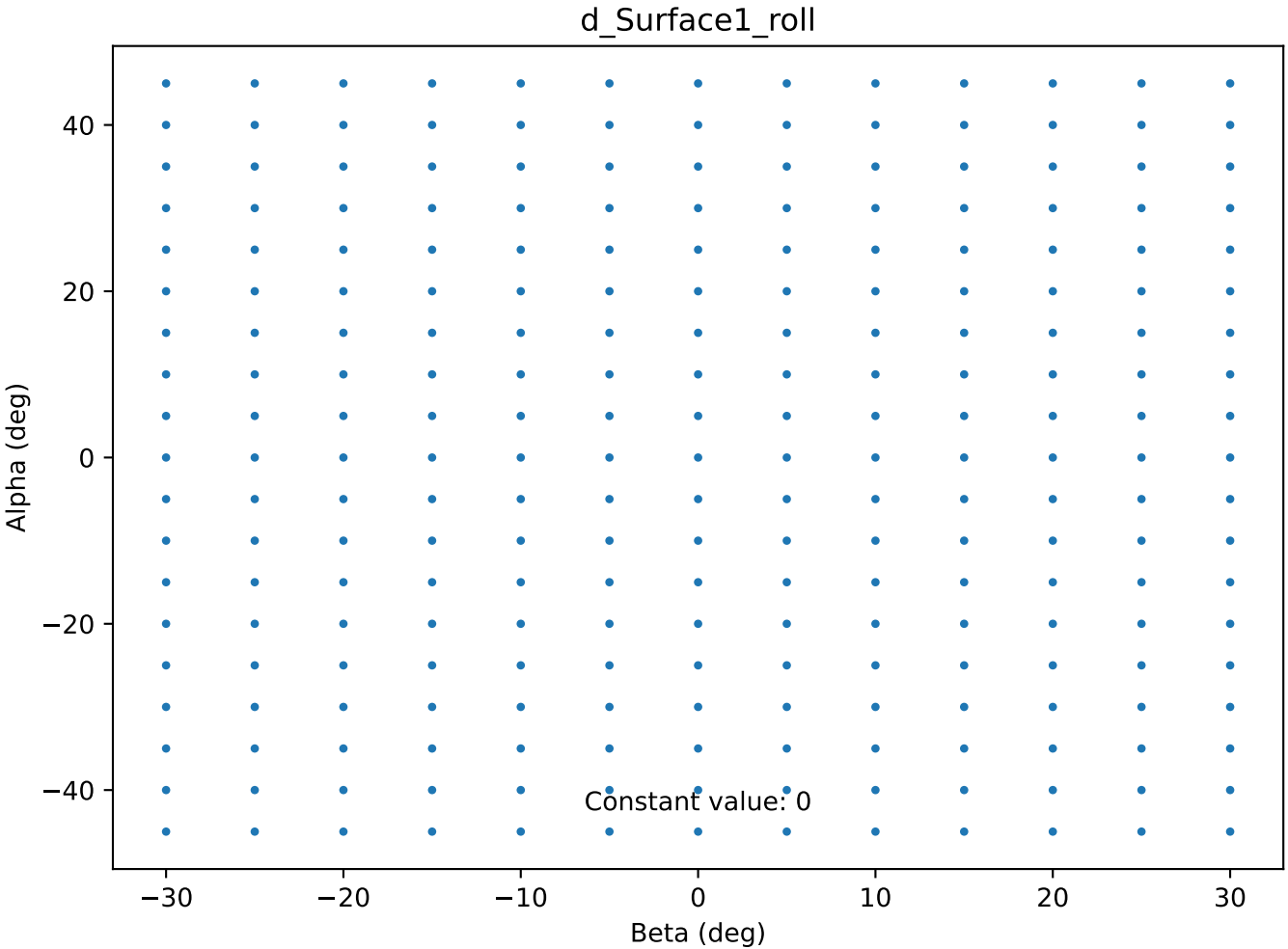

Dihedral effect (Clb)
Alpha (deg)
Beta (deg)
Dihedral effect (Clb)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.096
0.072
0.048
0.024
0.000
−0.024
−0.048
−0.072
−0.096
−0.120

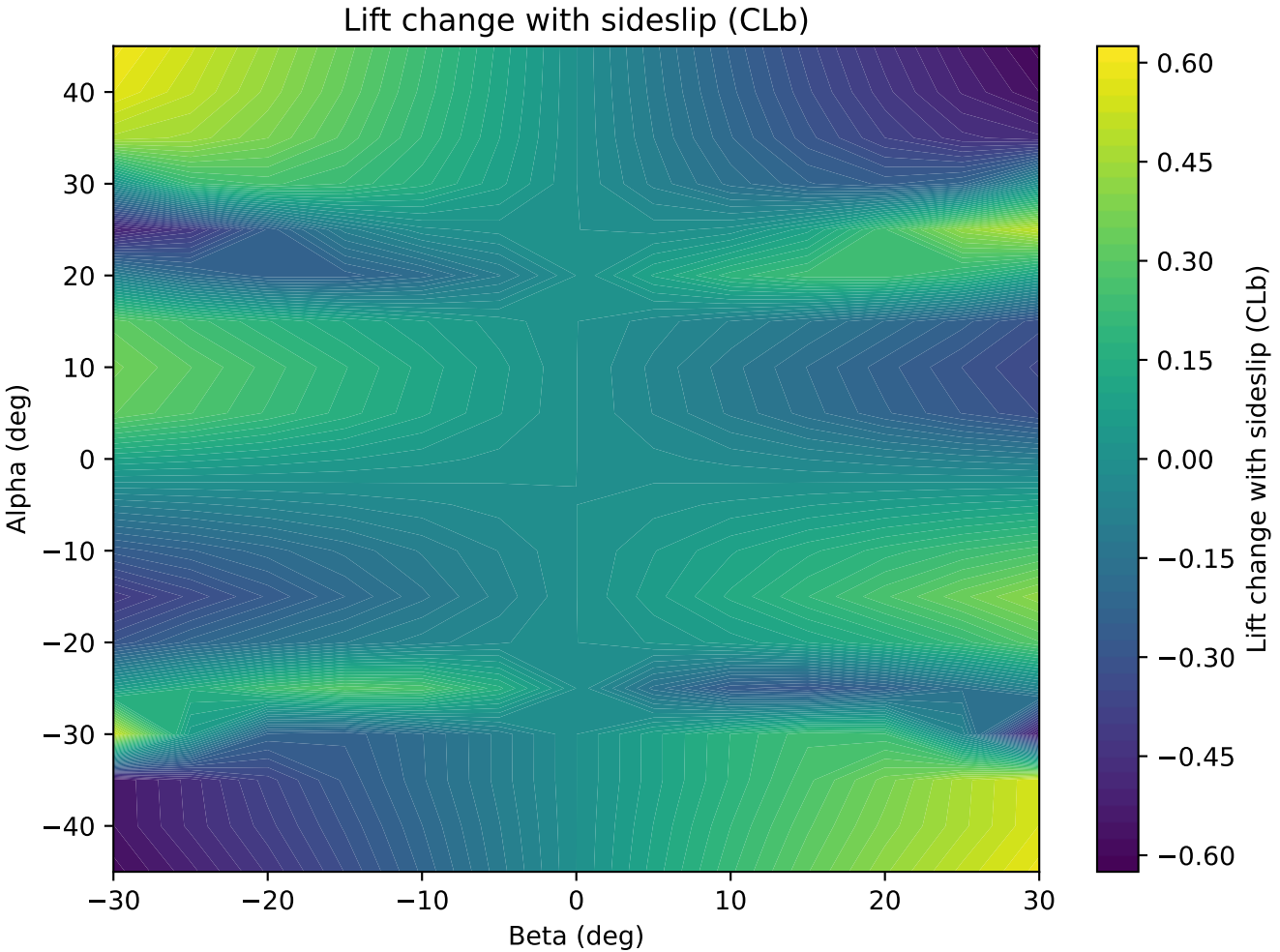

Roll damping (Clp)
Alpha (deg)
Beta (deg)
Roll damping (Clp)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.18
0.09
0.00
−0.09
−0.18
−0.27
−0.36
−0.45
−0.54

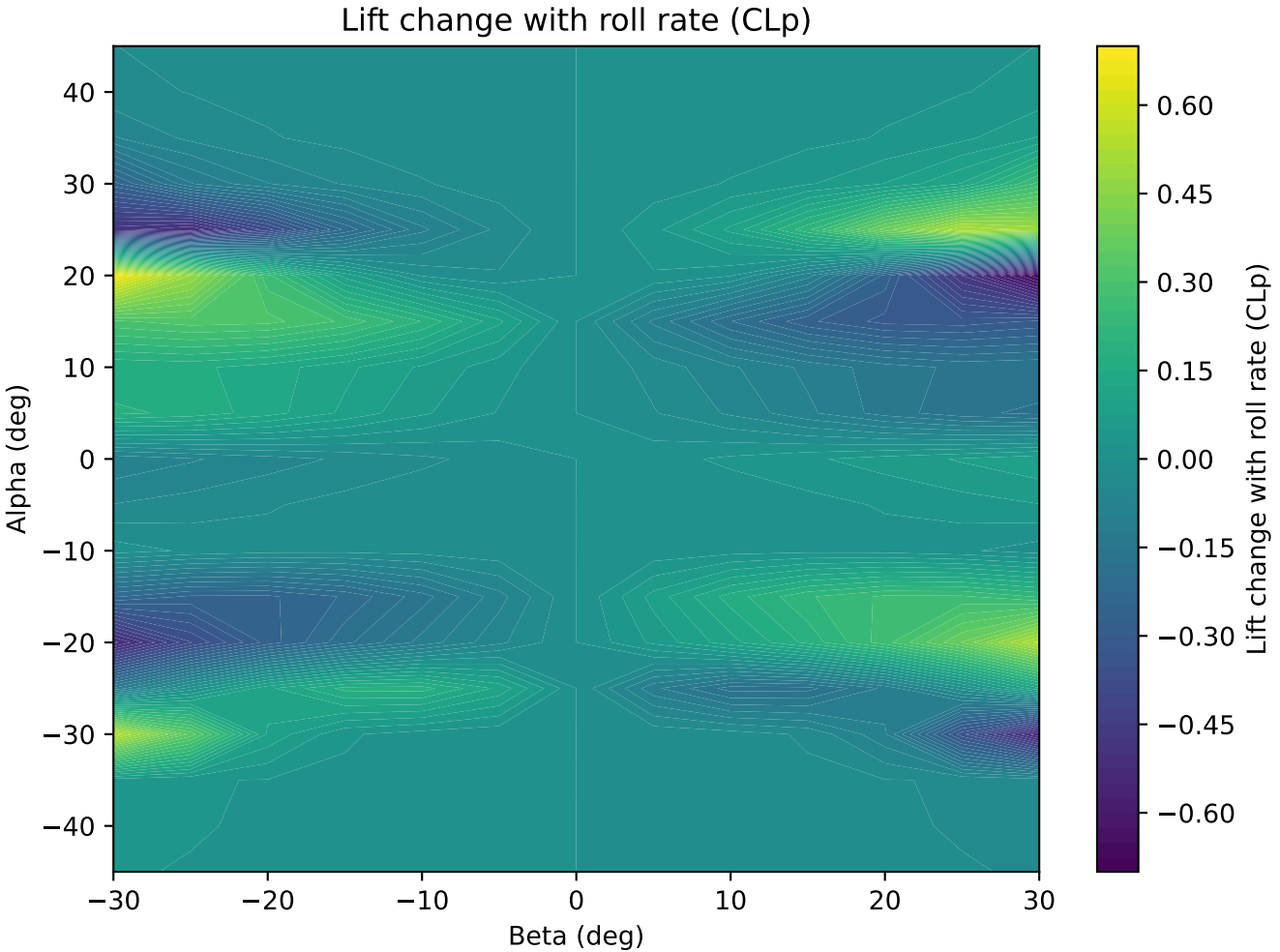

Pitch damping (Cmq)
Alpha (deg)
Beta (deg)
Pitch damping (Cmq)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.12
0.06
0.00
−0.06
−0.12
−0.18
−0.24
−0.30

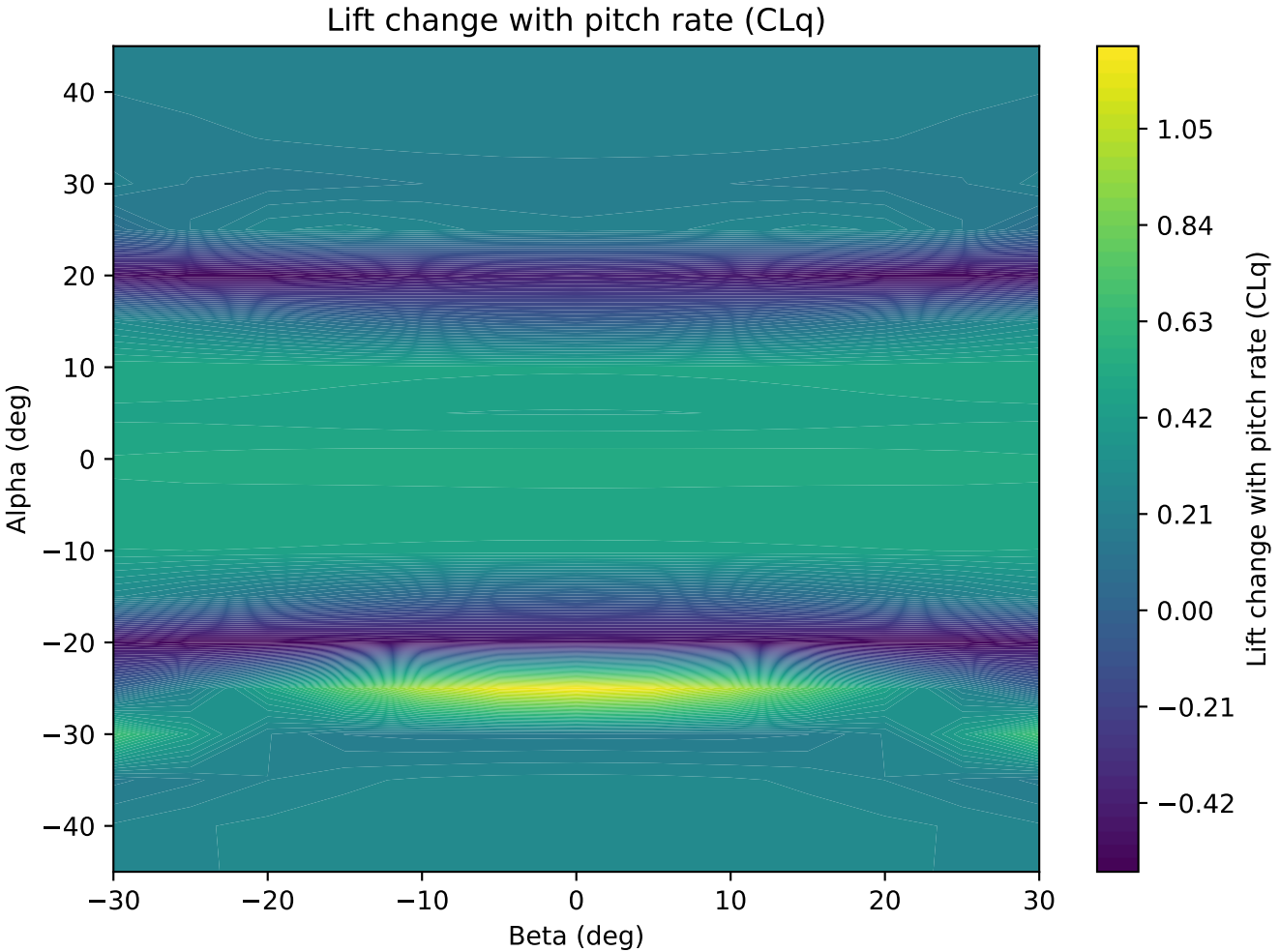

Yaw damping (Cnr)
Alpha (deg)
Beta (deg)
Yaw damping (Cnr)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.120
0.105
0.090
0.075
0.060
0.045
0.030
0.015
0.000

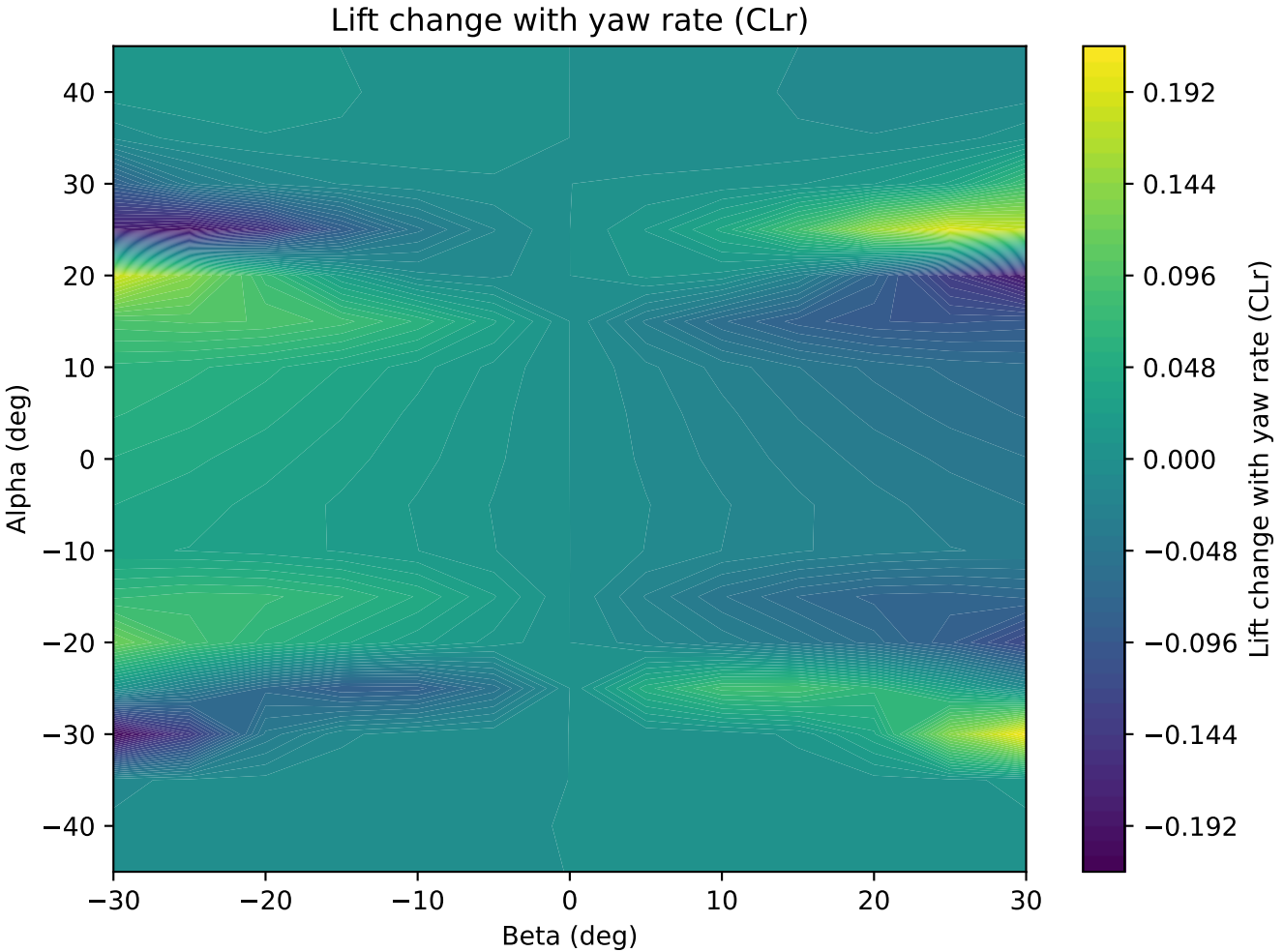

d_Surface1_pitch
Alpha (deg)
Beta (deg)
Constant value: 0
40
20
0
−20
−40
−30
−20
−10
0
10
20
30

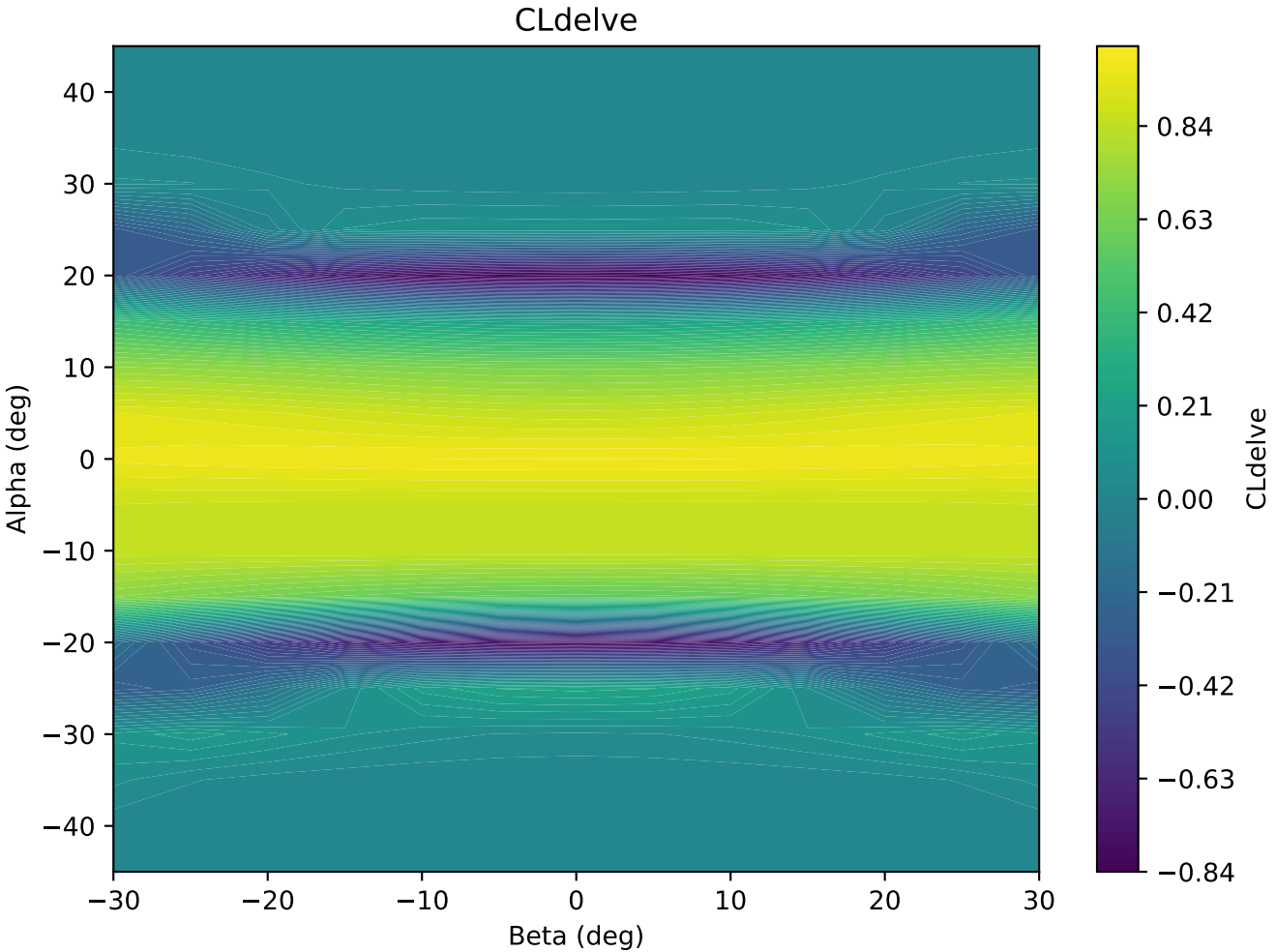

d_Surface1_roll
Alpha (deg)
Beta (deg)
40
20
0
−20
−40
−30
−20
−10
0
10
20
30
Constant value: 0

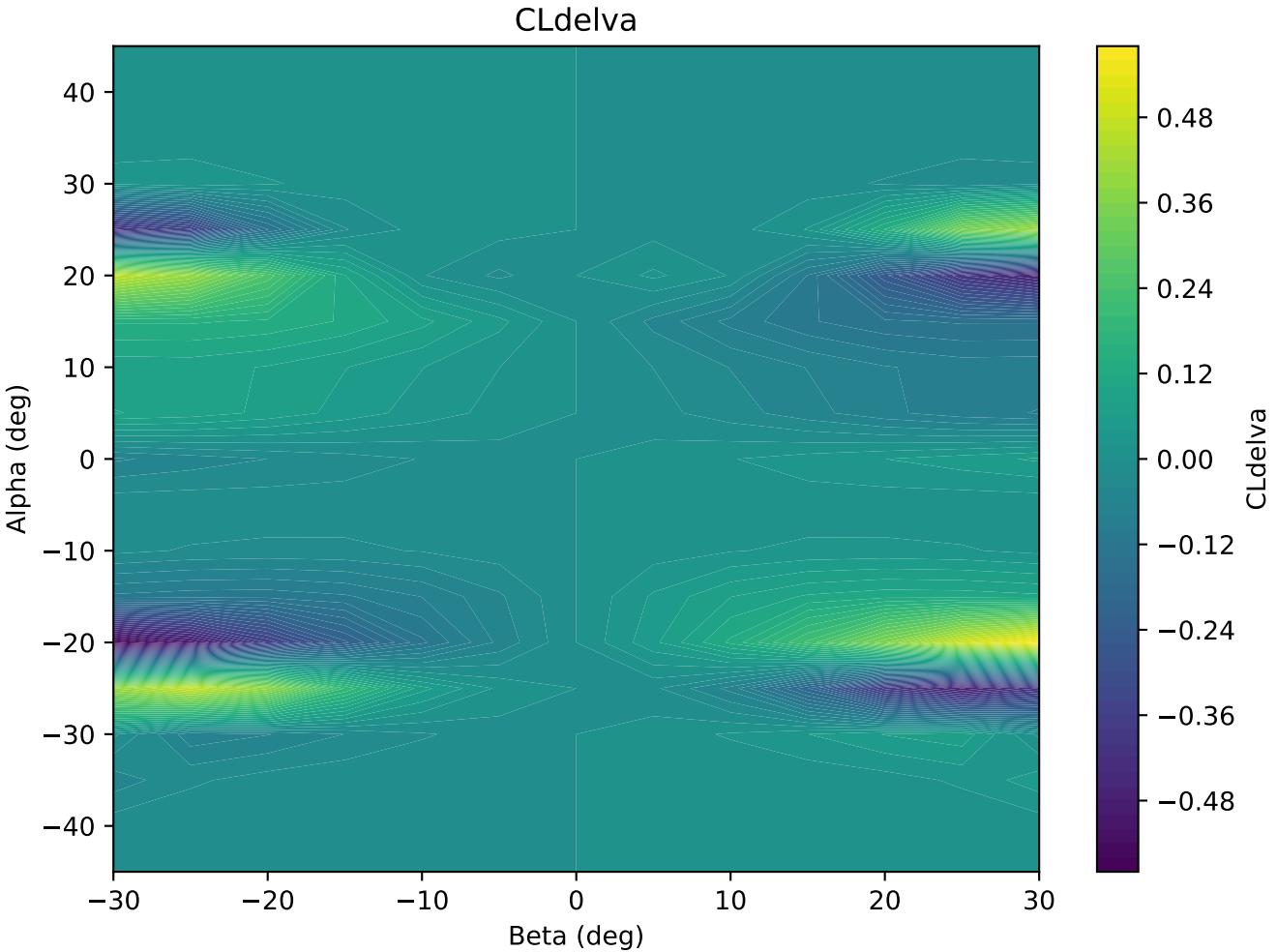

Lift change with sideslip (CLb)
Alpha (deg)
Beta (deg)
Lift change with sideslip (CLb)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30
−0.45
−0.60

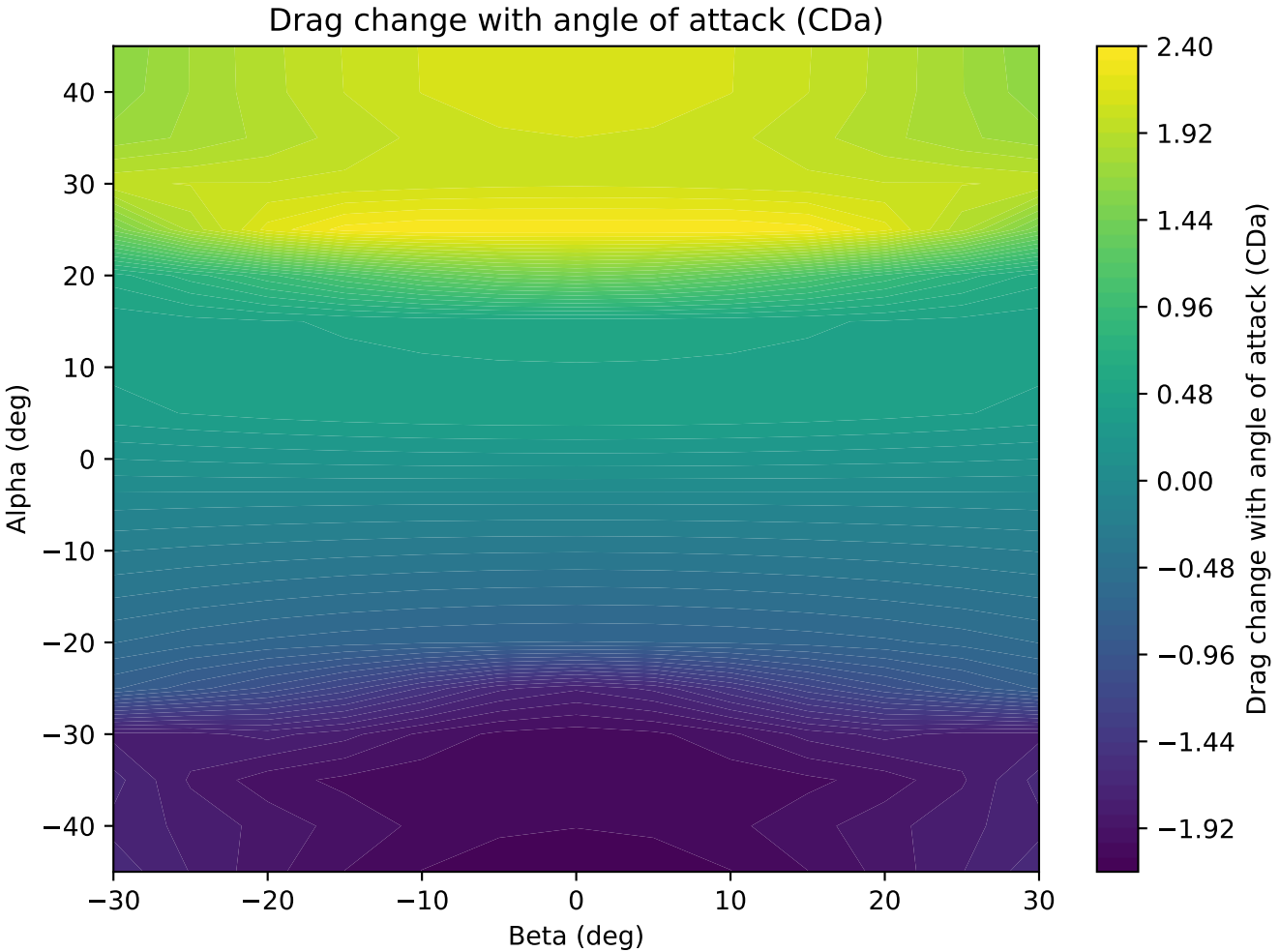

Lift change with roll rate (CLp)
Alpha (deg)
Beta (deg)
Lift change with roll rate (CLp)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30
−0.45
−0.60

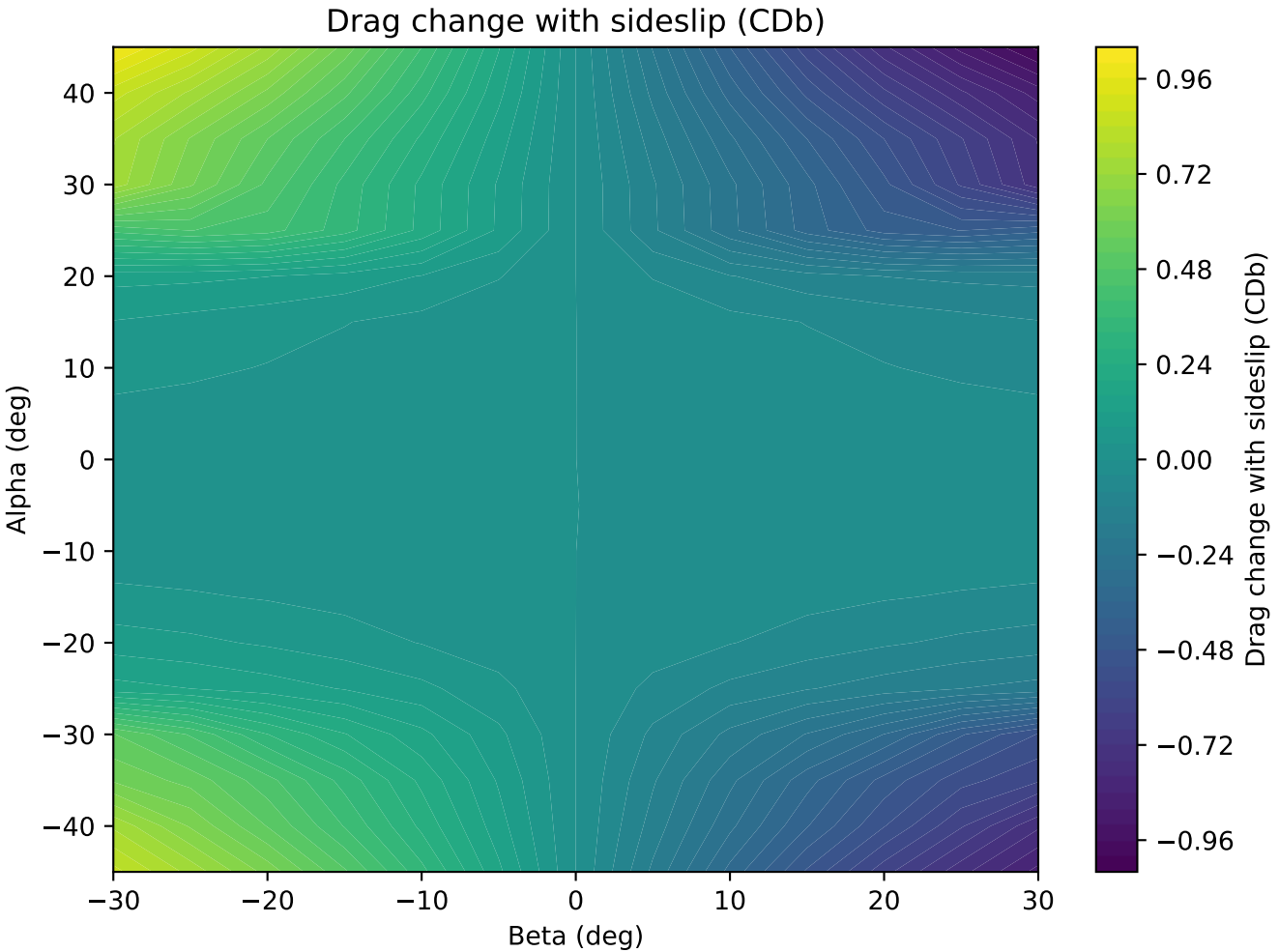

Lift change with pitch rate (CLq)
Alpha (deg)
Beta (deg)
Lift change with pitch rate (CLq)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
1.05
0.84
0.63
0.42
0.21
0.00
−0.21
−0.42

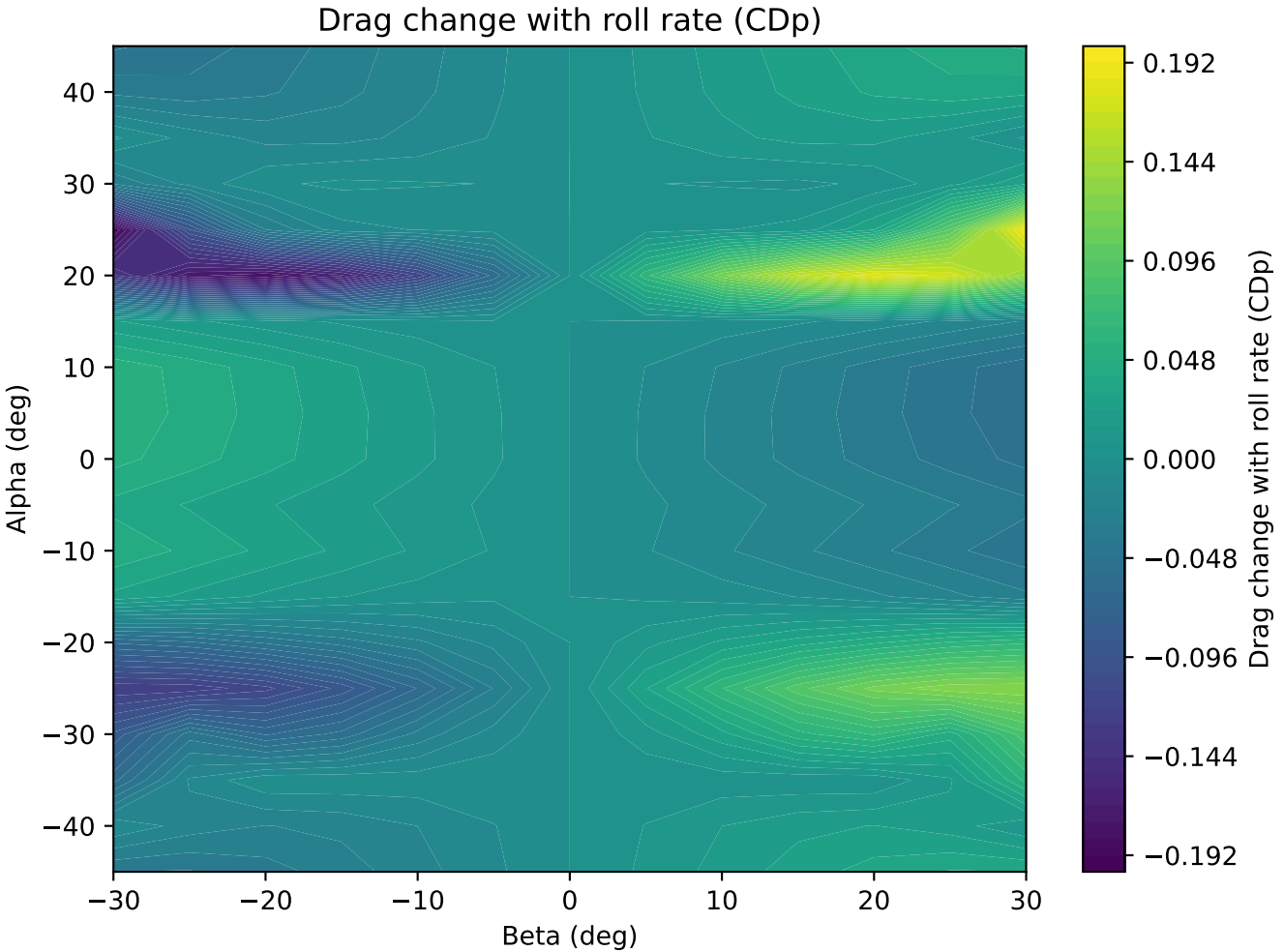

Lift change with yaw rate (CLr)
Alpha (deg)
Beta (deg)
Lift change with yaw rate (CLr)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.192
0.144
0.096
0.048
0.000
−0.048
−0.096
−0.144
−0.192

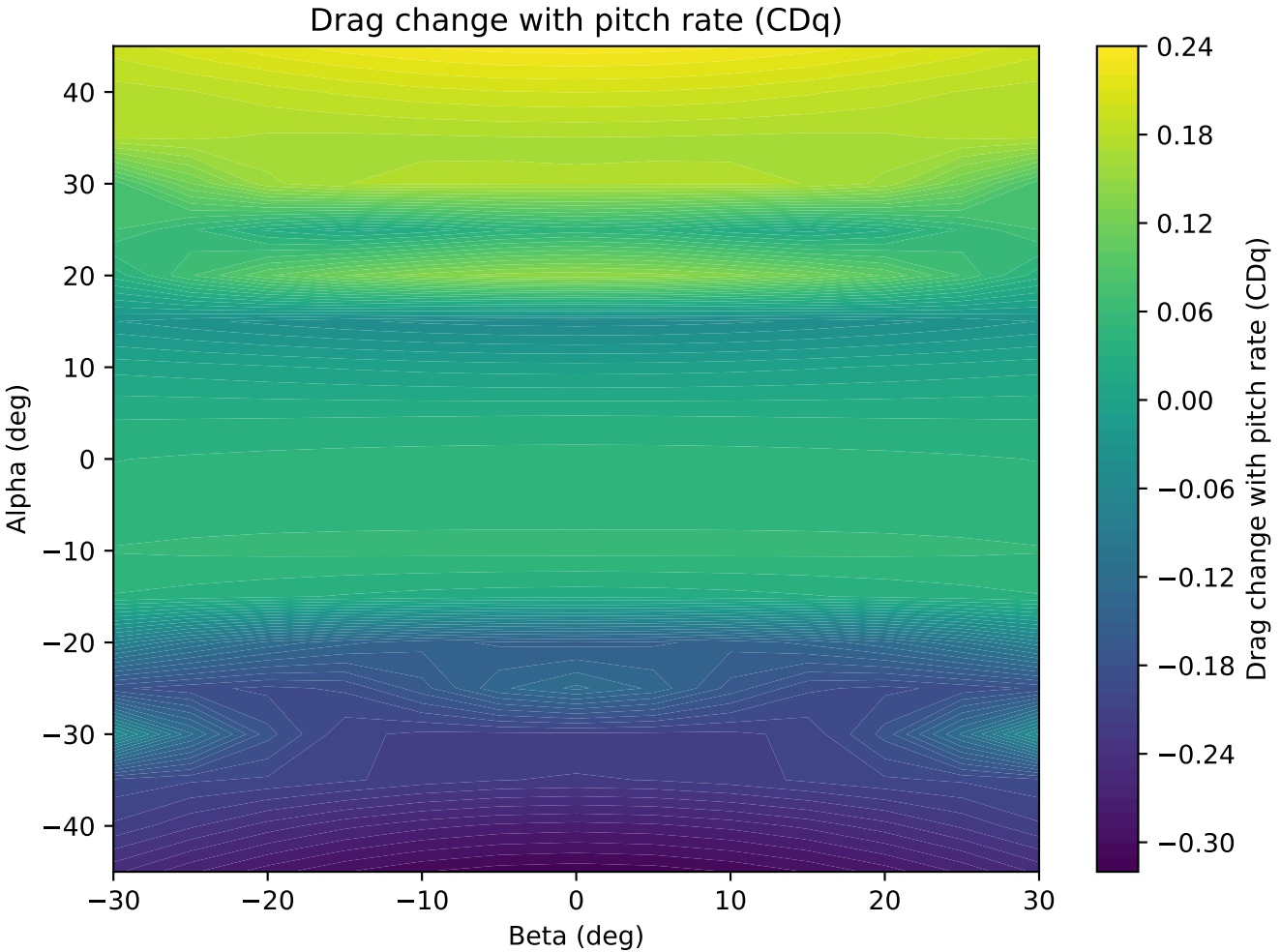

CLdelve
Alpha (deg)
Beta (deg)
CLdelve
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.84
0.63
0.42
0.21
0.00
−0.21
−0.42
−0.63
−0.84

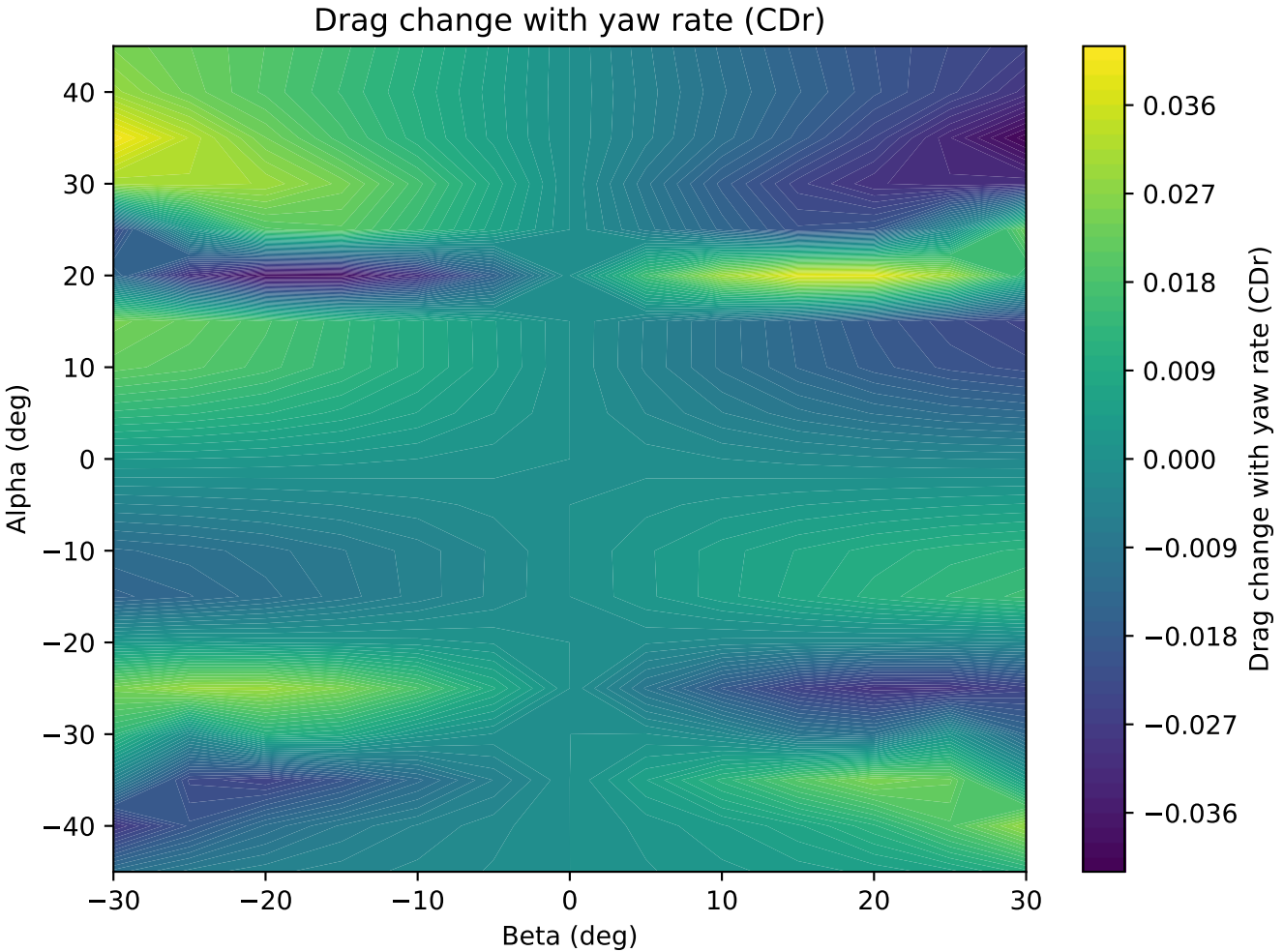

CLdelva
Alpha (deg)
Beta (deg)
CLdelva
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.48
0.36
0.24
0.12
0.00
−0.12
−0.24
−0.36
−0.48

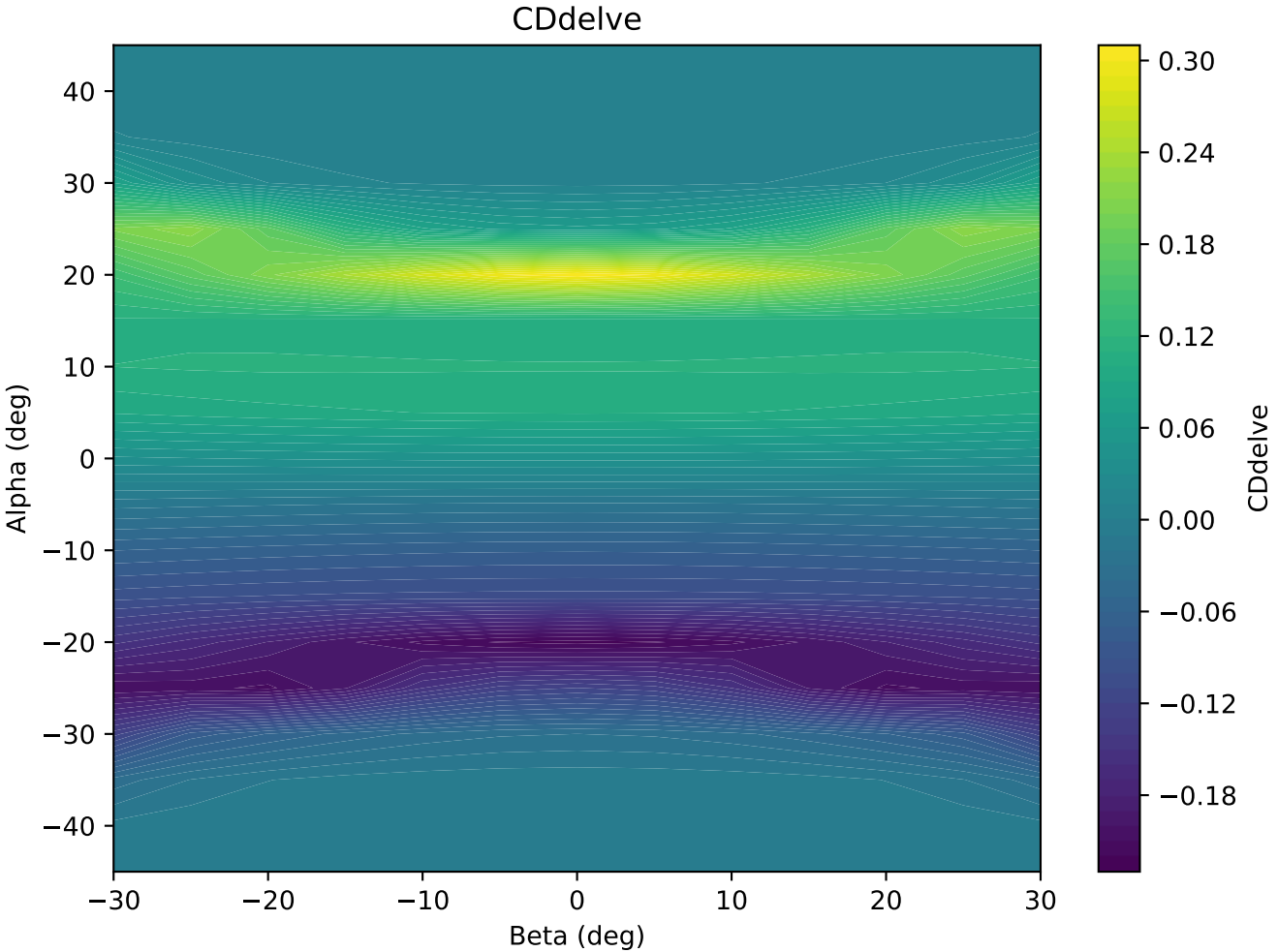

Drag change with angle of attack (CDa)
Alpha (deg)
Beta (deg)
Drag change with angle of attack (CDa)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
2.40
1.92
1.44
0.96
0.48
0.00
−0.48
−0.96
−1.44
−1.92

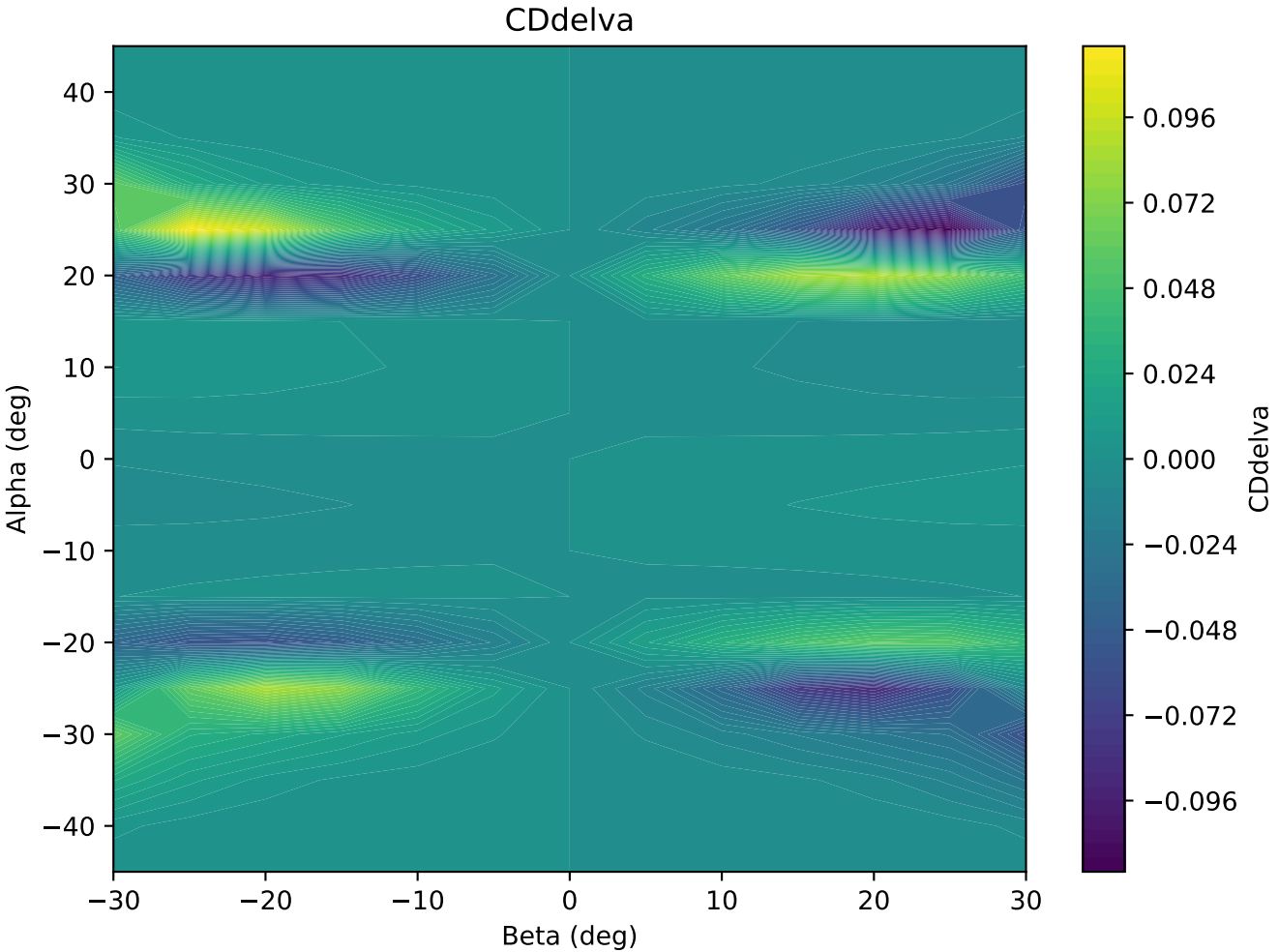

Drag change with sideslip (CDb)
Alpha (deg)
Beta (deg)
Drag change with sideslip (CDb)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.96
0.72
0.48
0.24
0.00
−0.24
−0.48
−0.72
−0.96

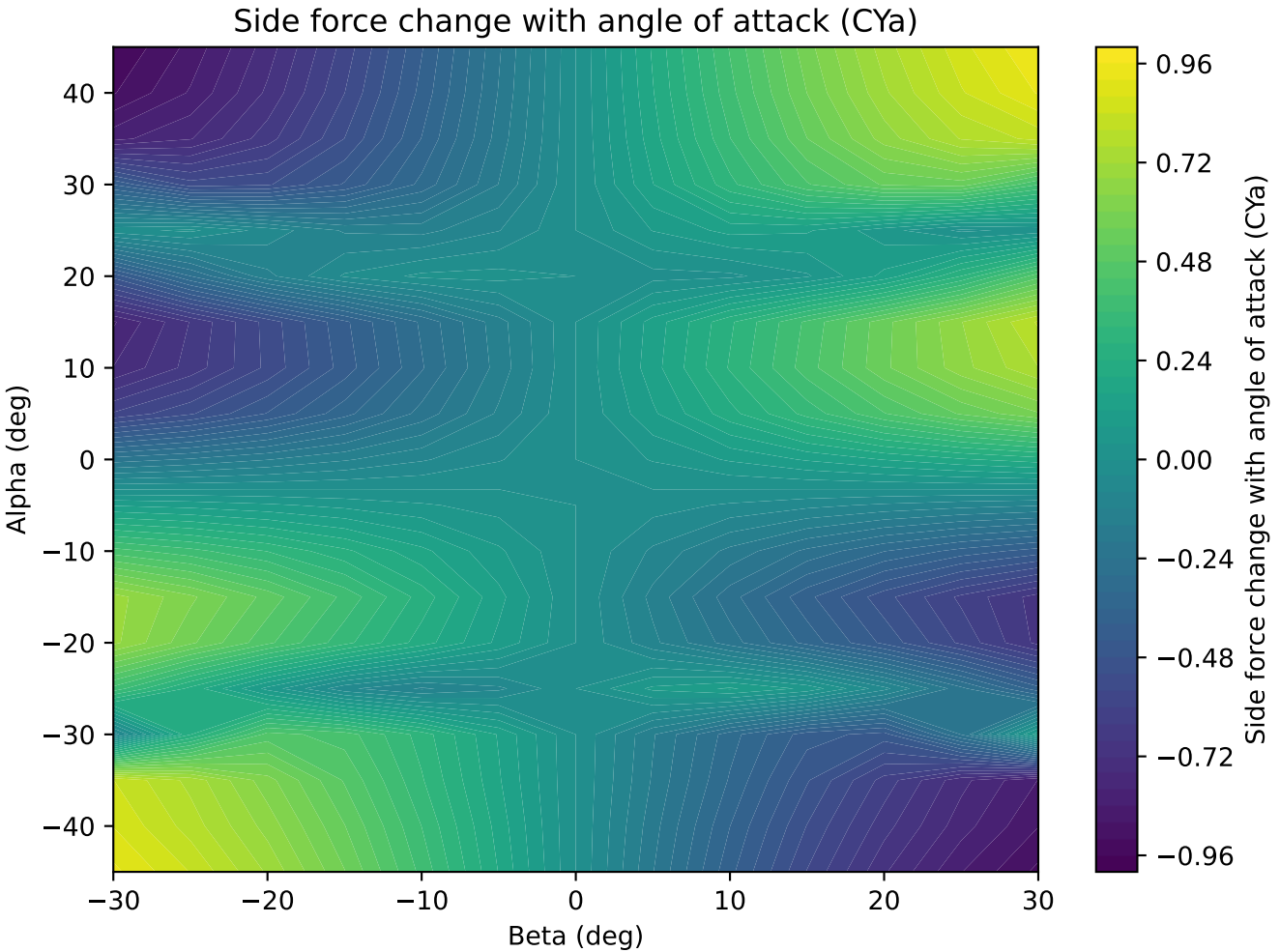

Drag change with roll rate (CDp)
Alpha (deg)
Beta (deg)
Drag change with roll rate (CDp)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.192
0.144
0.096
0.048
0.000
−0.048
−0.096
−0.144
−0.192

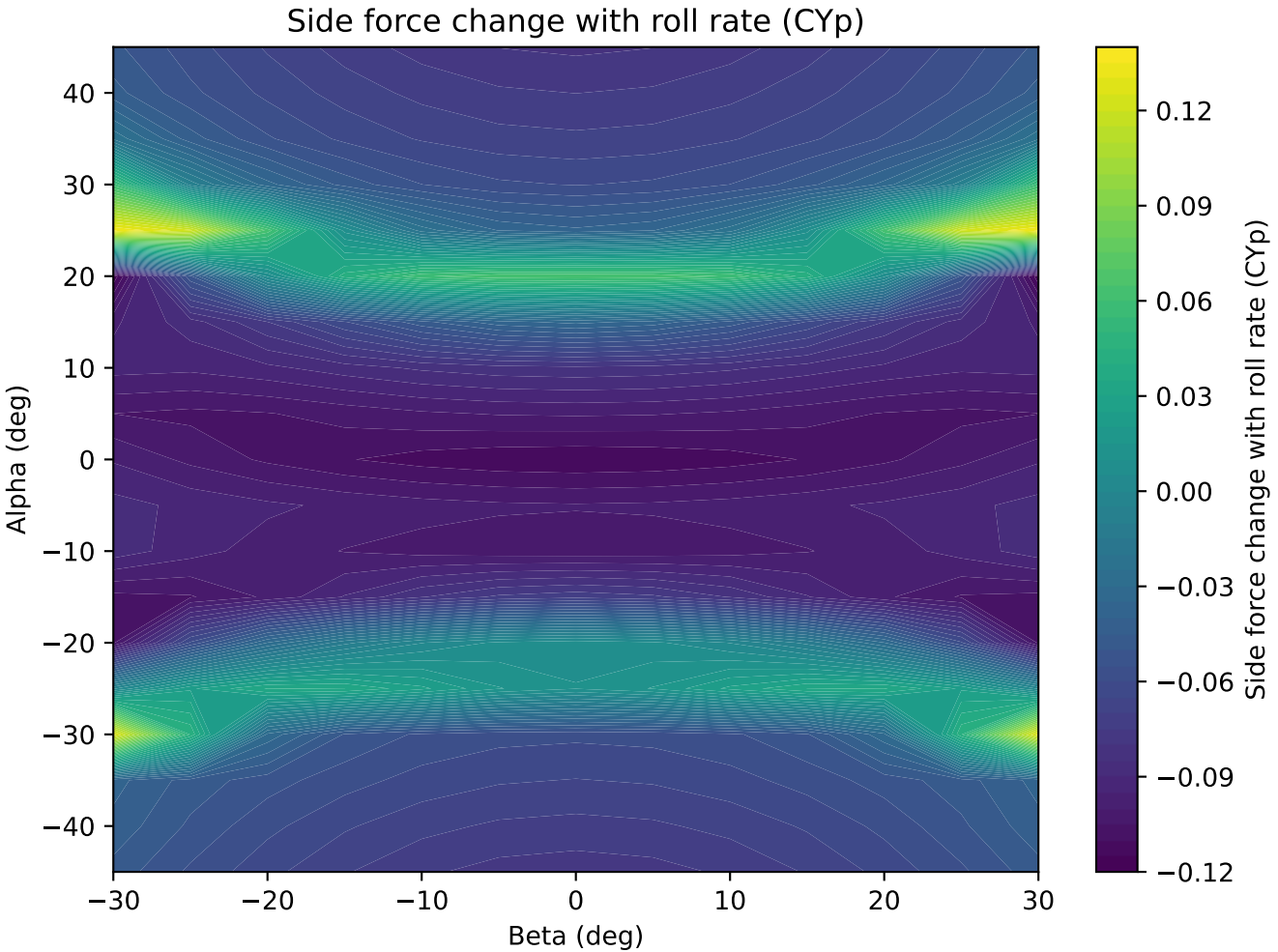

Drag change with pitch rate (CDq)
Alpha (deg)
Beta (deg)
Drag change with pitch rate (CDq)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.24
0.18
0.12
0.06
0.00
−0.06
−0.12
−0.18
−0.24
−0.30

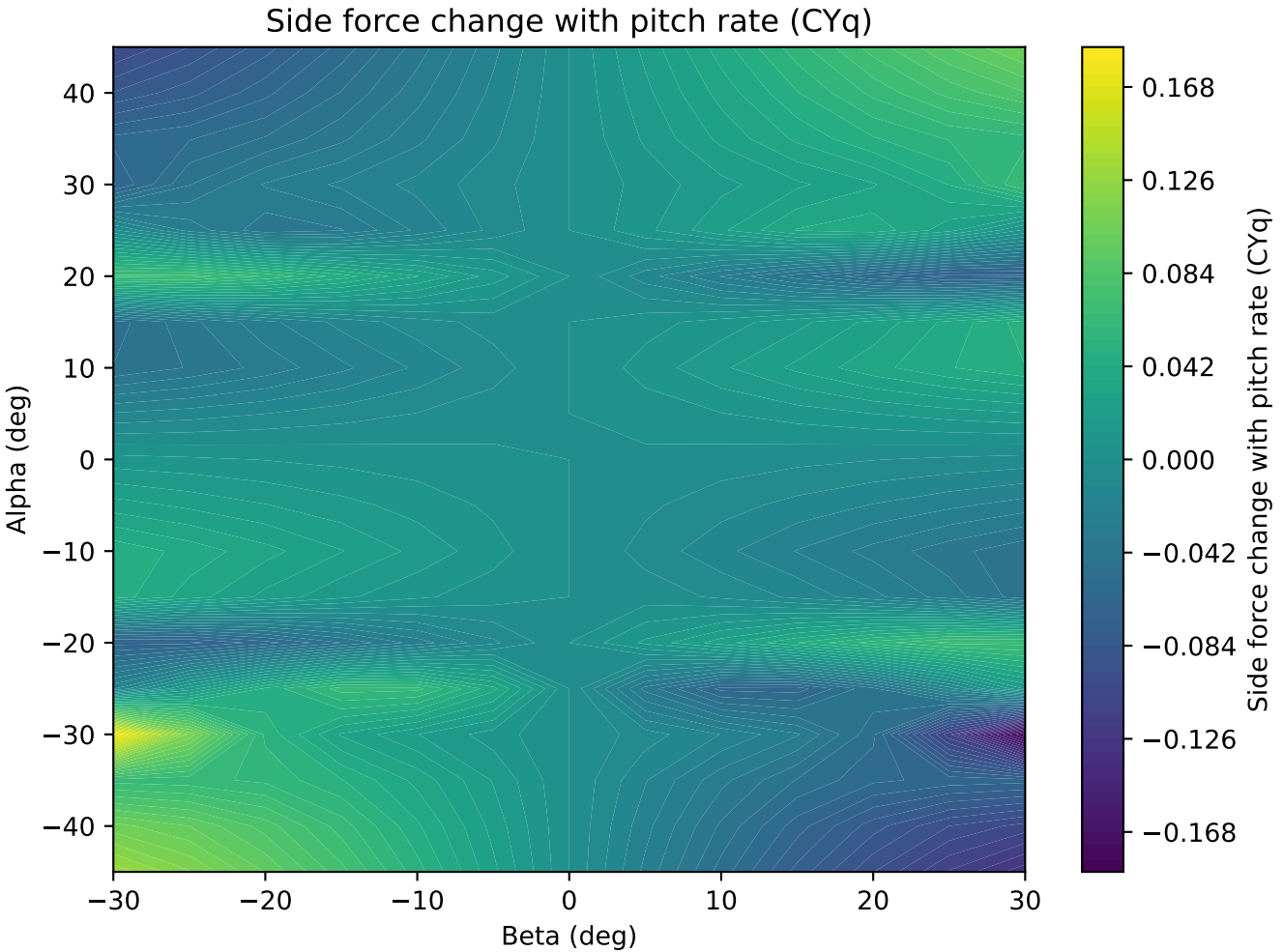

Drag change with yaw rate (CDr)
Alpha (deg)
Beta (deg)
Drag change with yaw rate (CDr)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.036
0.027
0.018
0.009
0.000
−0.009
−0.018
−0.027
−0.036

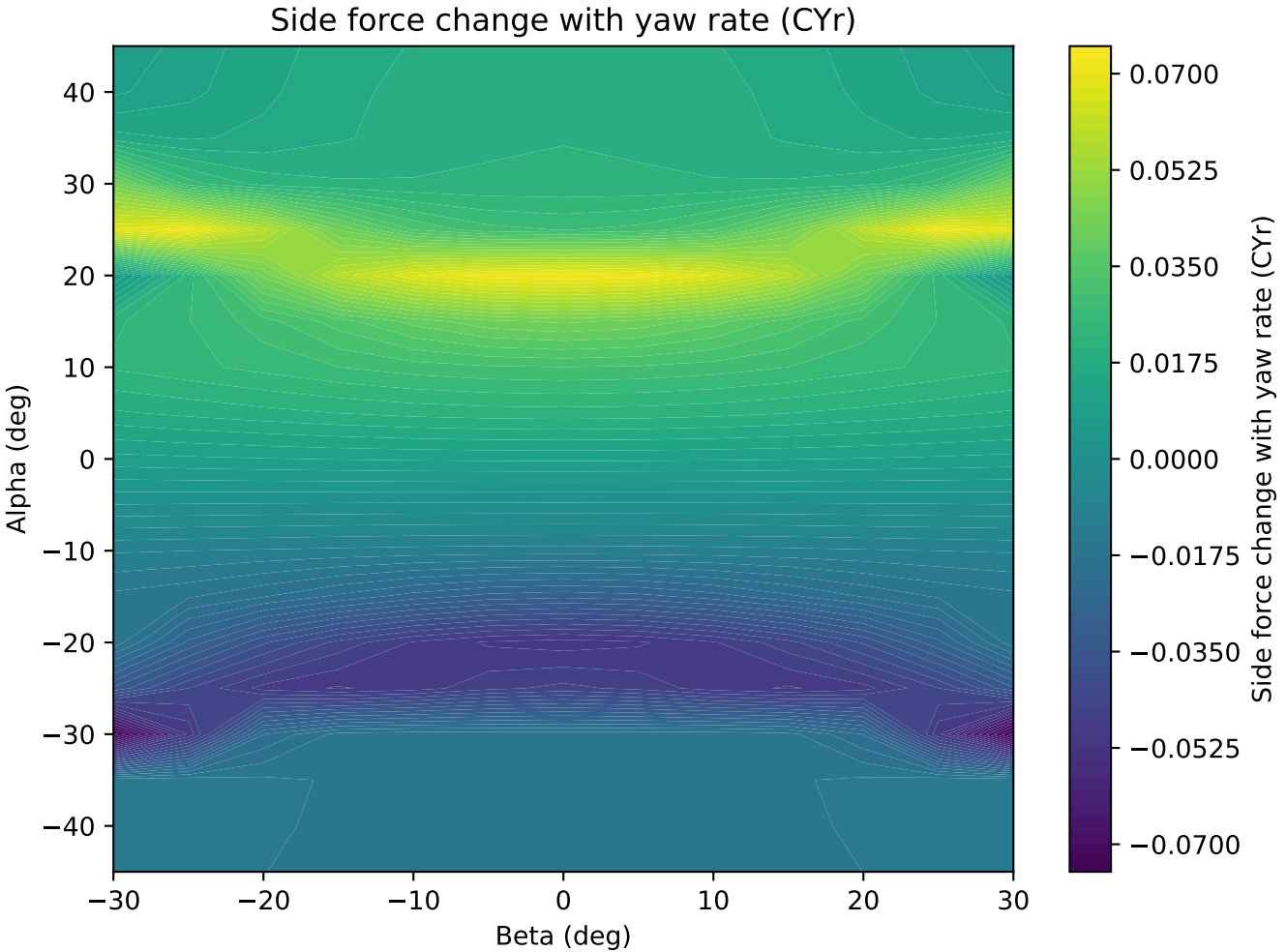

CDdelve
Alpha (deg)
Beta (deg)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.30
0.24
0.18
0.12
0.06
0.00
−0.06
−0.12
−0.18
CDdelve

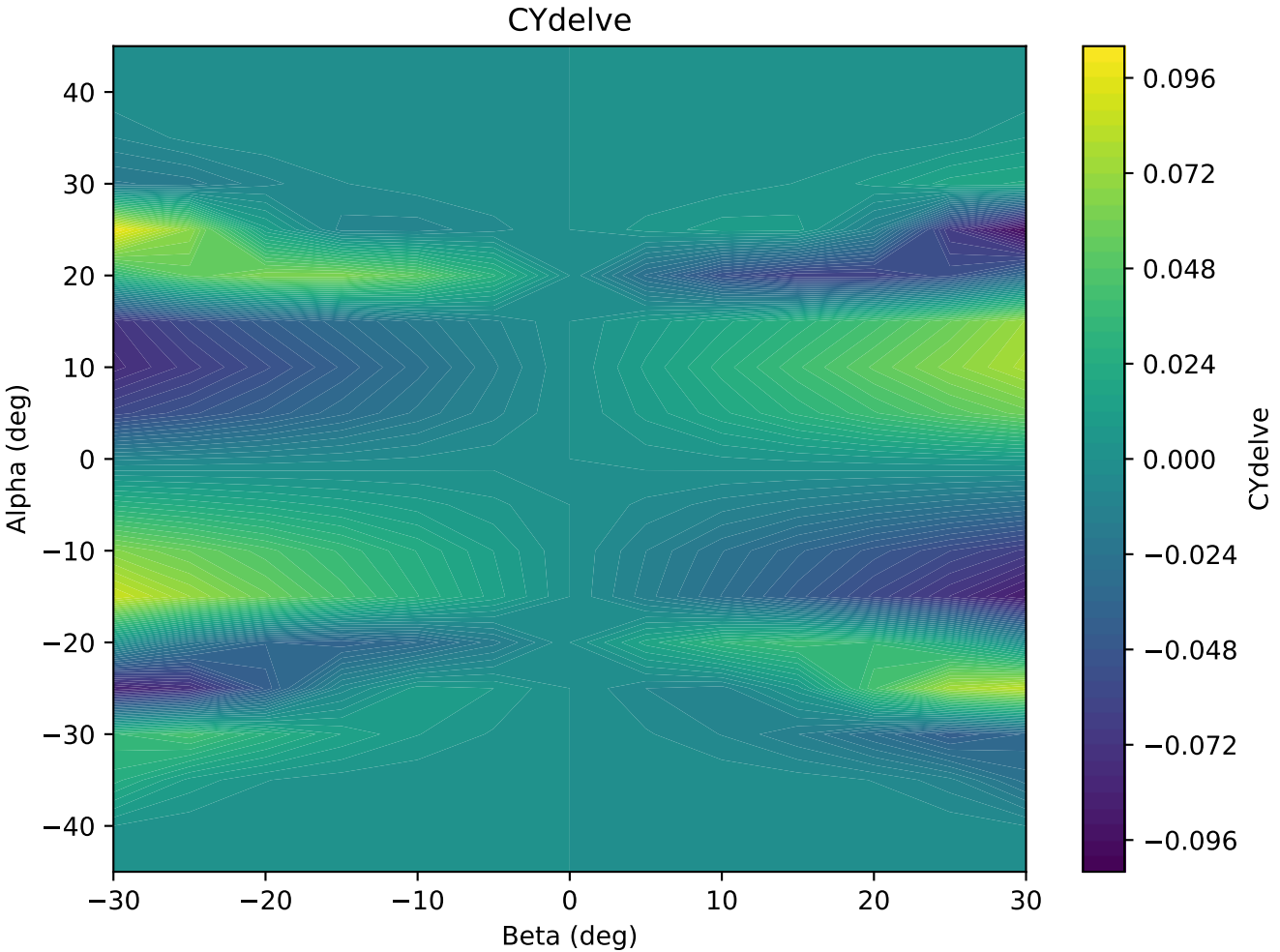

CDdelva
Alpha (deg)
Beta (deg)
CDdelva
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.096
0.072
0.048
0.024
0.000
−0.024
−0.048
−0.072
−0.096

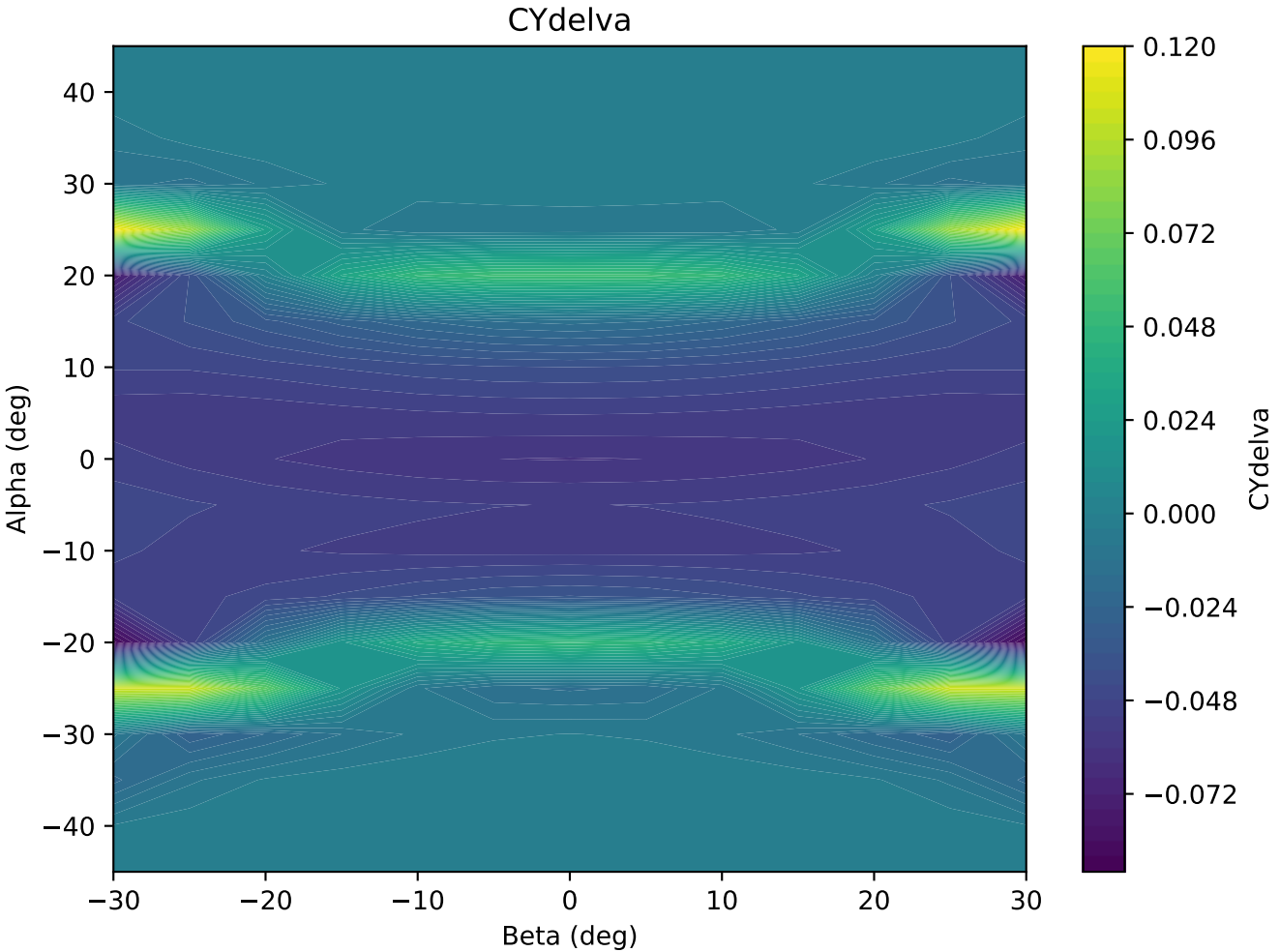

Side force change with angle of attack (CYa)
Alpha (deg)
Beta (deg)
Side force change with angle of attack (CYa)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.96
0.72
0.48
0.24
0.00
−0.24
−0.48
−0.72
−0.96

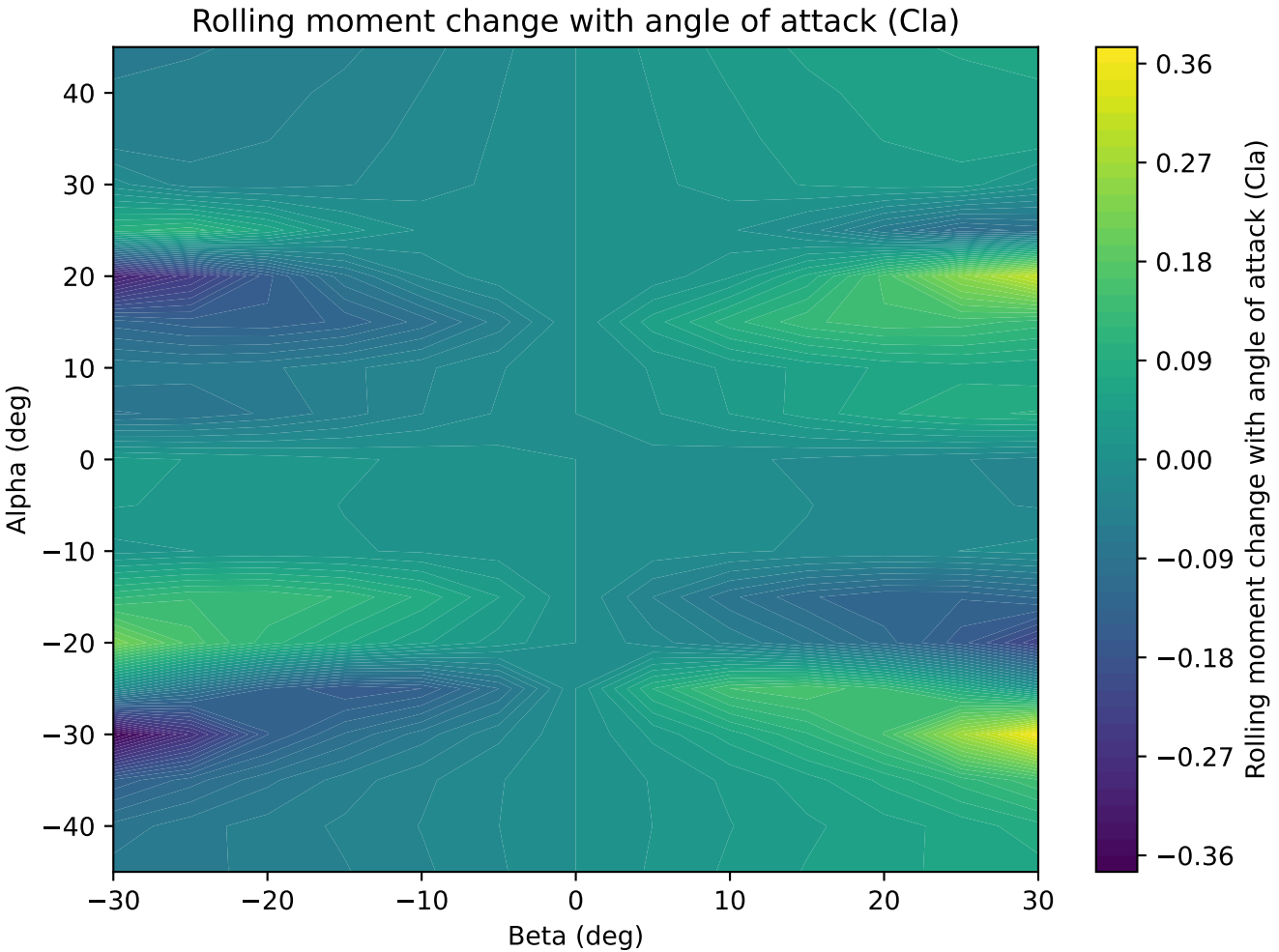

Side force change with roll rate (CYp)
Alpha (deg)
Beta (deg)
Side force change with roll rate (CYp)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.12
0.09
0.06
0.03
0.00
−0.03
−0.06
−0.09
−0.12

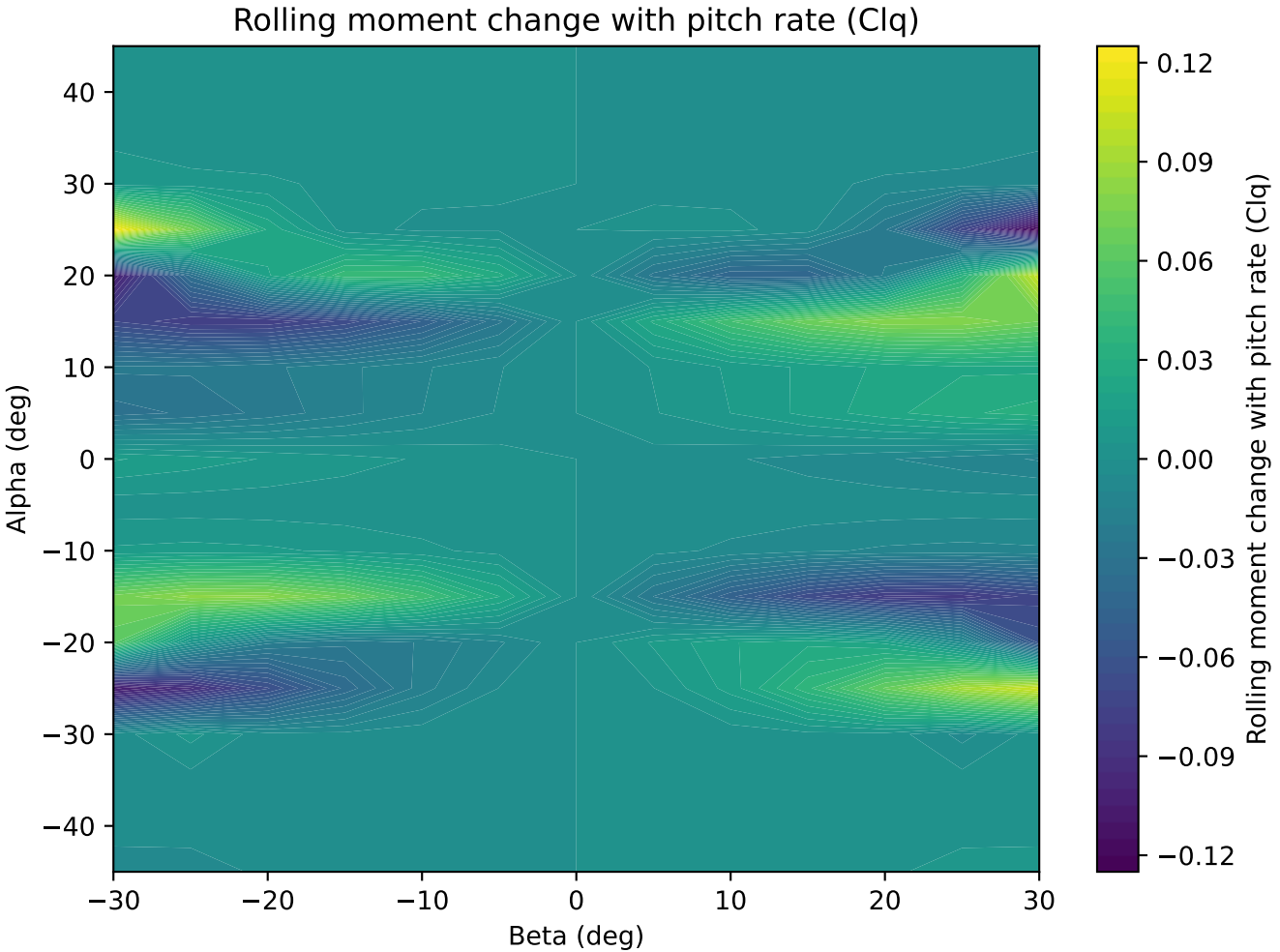

Side force change with pitch rate (CYq)
Alpha (deg)
Beta (deg)
Side force change with pitch rate (CYq)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.168
0.126
0.084
0.042
0.000
−0.042
−0.084
−0.126
−0.168

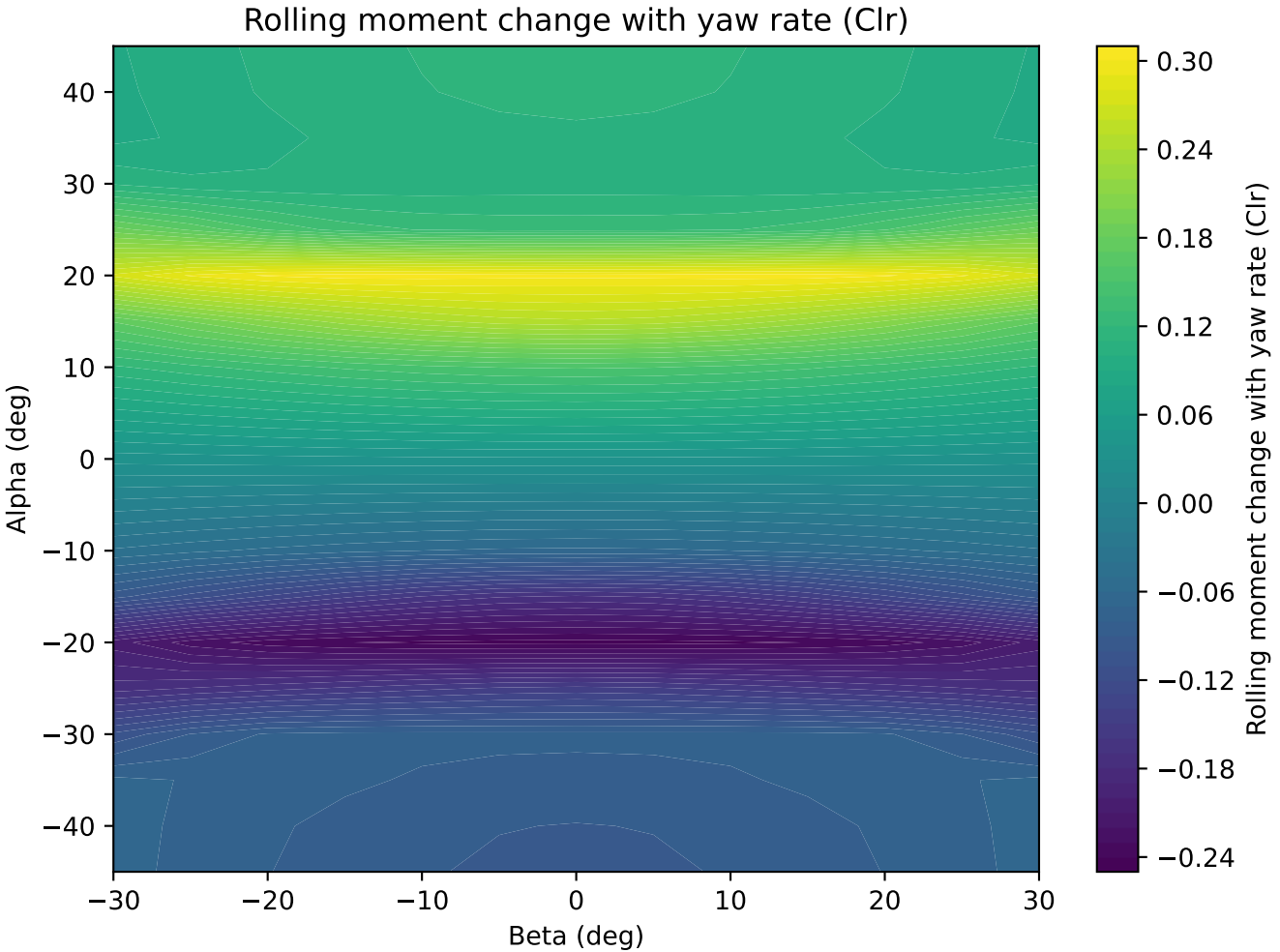

Side force change with yaw rate (CYr)
Alpha (deg)
Beta (deg)
Side force change with yaw rate (CYr)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.0700
0.0525
0.0350
0.0175
0.0000
−0.0175
−0.0350
−0.0525
−0.0700

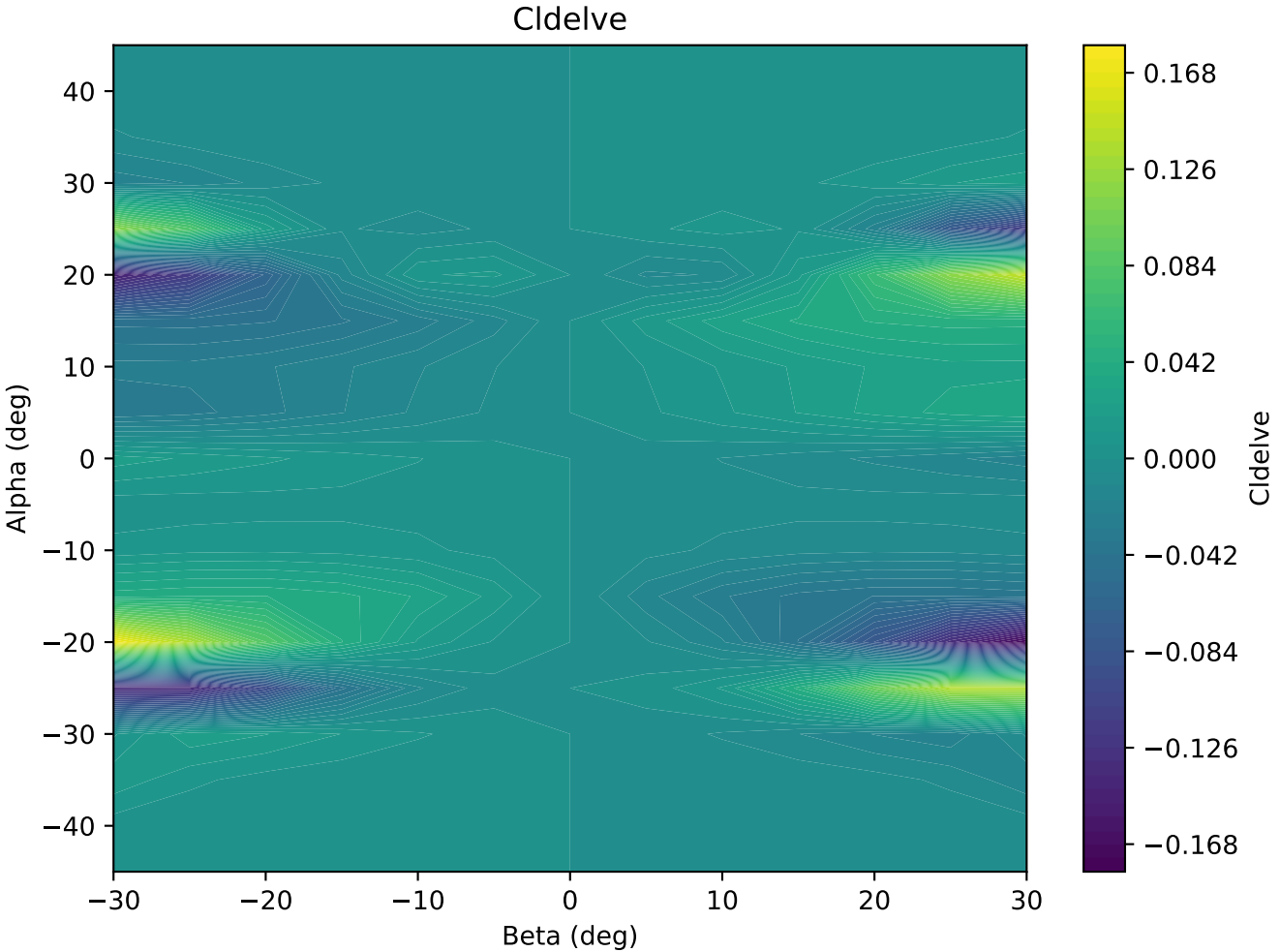

CYdelve
Alpha (deg)
Beta (deg)
CYdelve
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.096
0.072
0.048
0.024
0.000
−0.024
−0.048
−0.072
−0.096

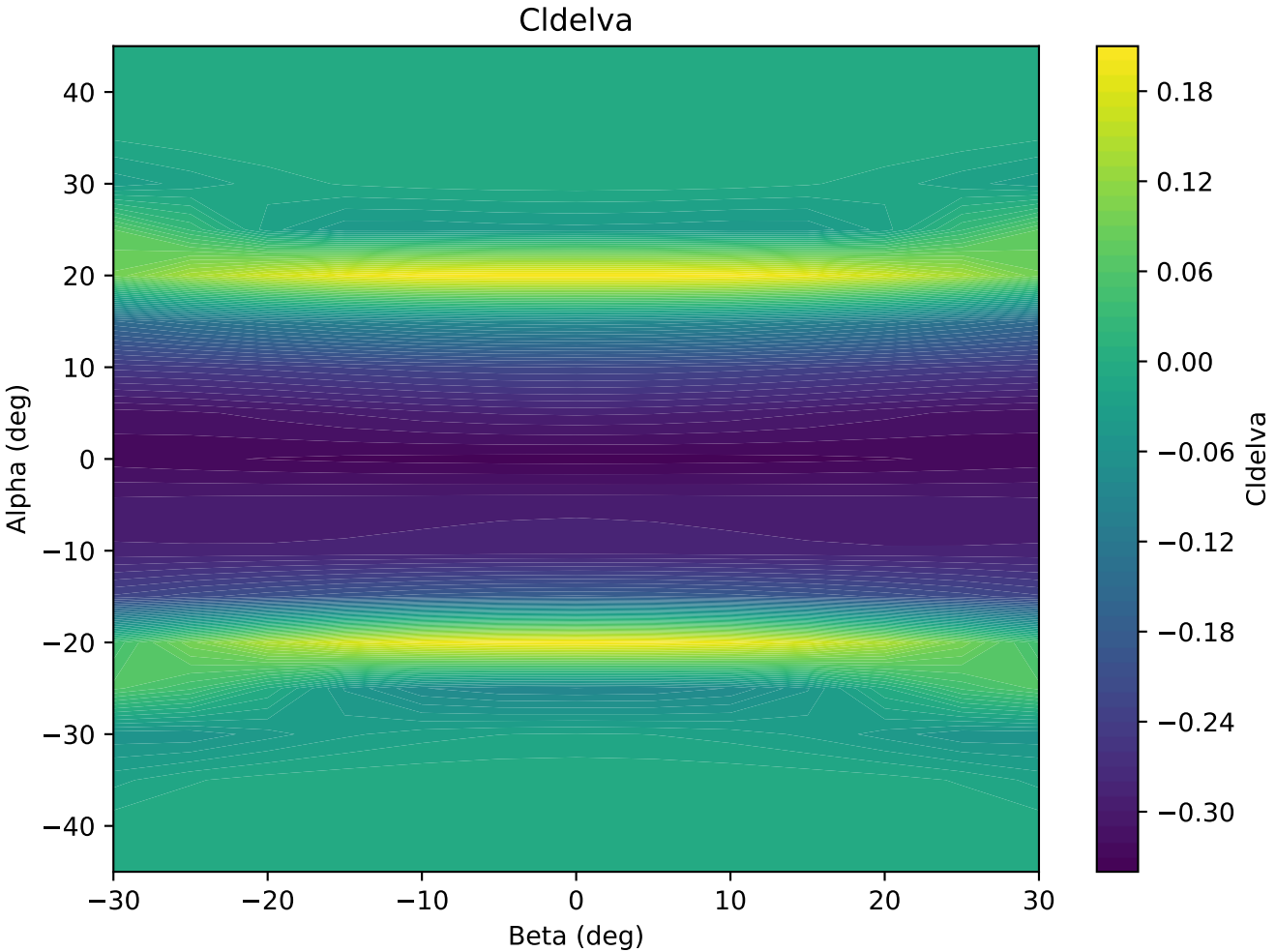

CYdelva
Alpha (deg)
Beta (deg)
CYdelva
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.120
0.096
0.072
0.048
0.024
0.000
−0.024
−0.048
−0.072

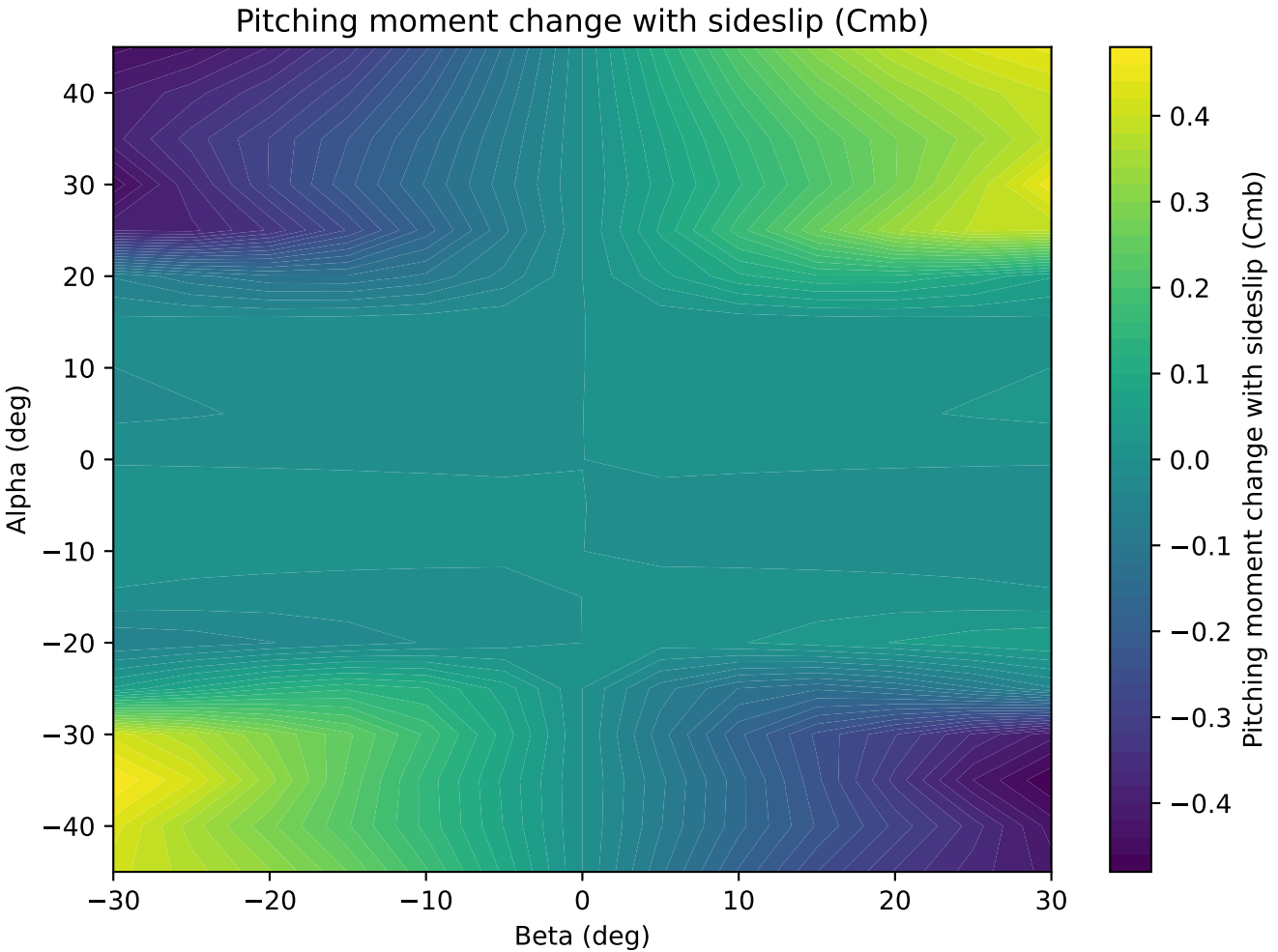

Rolling moment change with angle of attack (Cla)
Alpha (deg)
Beta (deg)
Rolling moment change with angle of attack (Cla)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.36
0.27
0.18
0.09
0.00
−0.09
−0.18
−0.27
−0.36

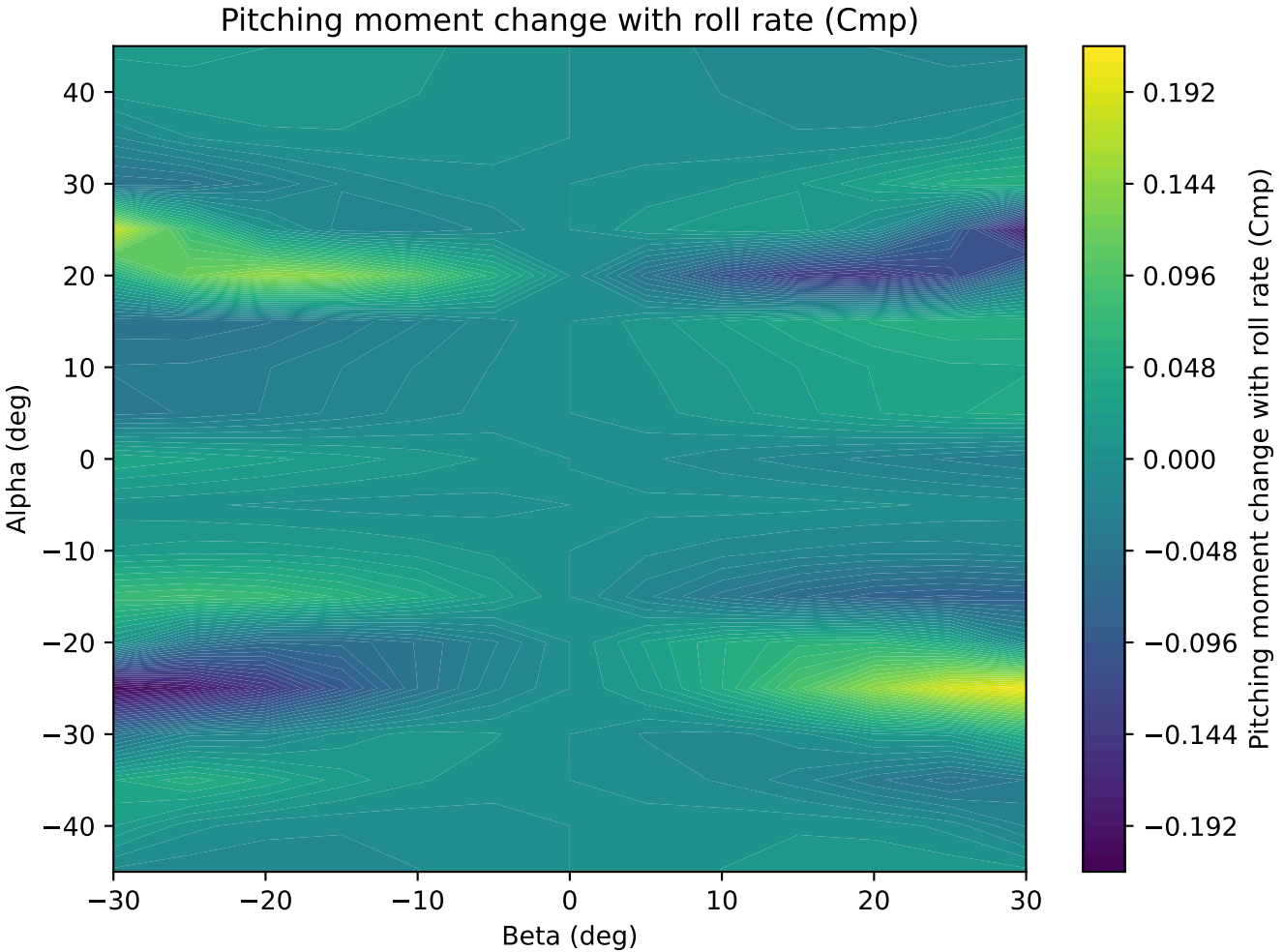

Rolling moment change with pitch rate (Clq)
Alpha (deg)
Beta (deg)
Rolling moment change with pitch rate (Clq)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.12
0.09
0.06
0.03
0.00
−0.03
−0.06
−0.09
−0.12

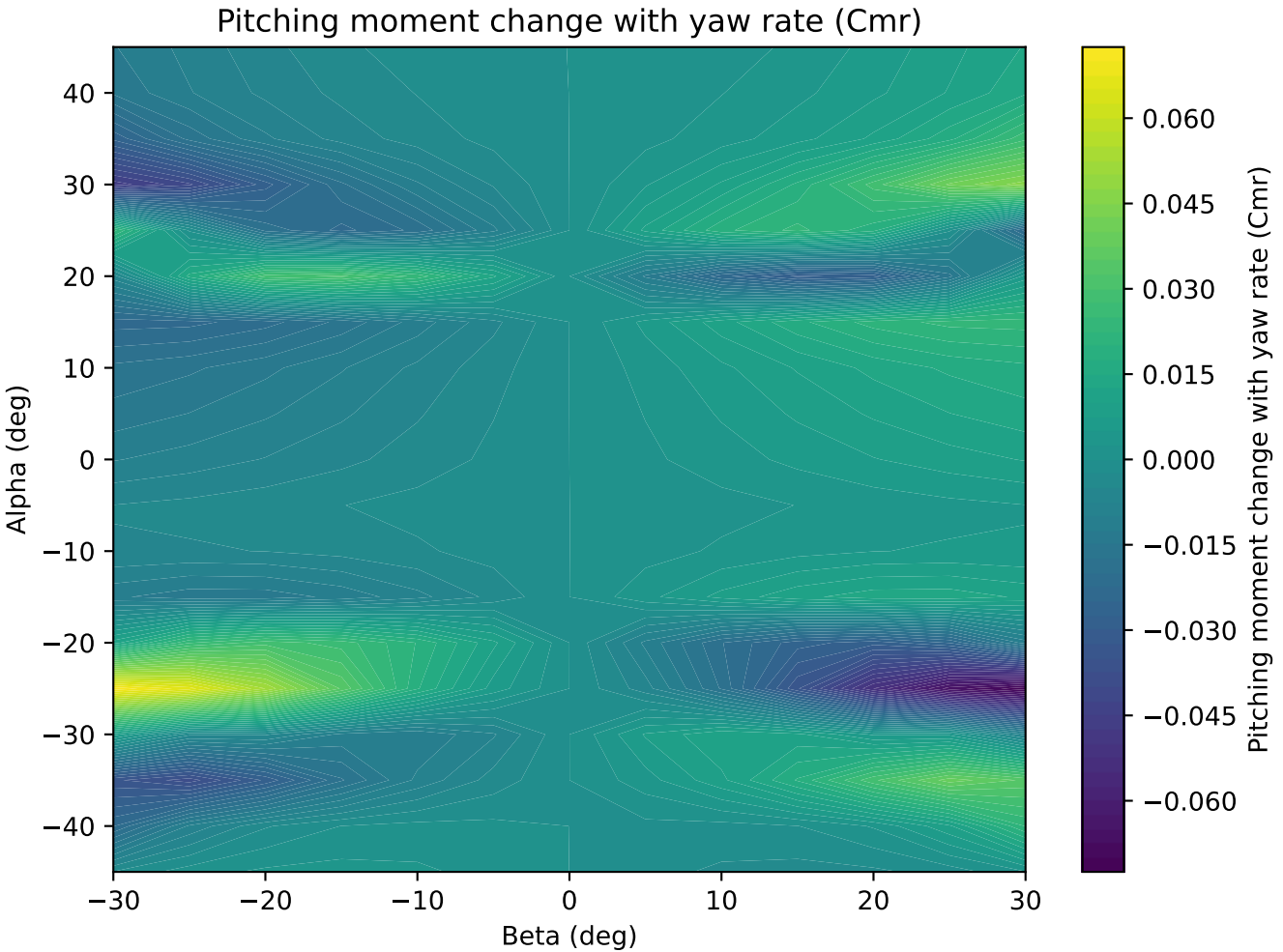

Rolling moment change with yaw rate (Clr)
Alpha (deg)
Beta (deg)
Rolling moment change with yaw rate (Clr)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.30
0.24
0.18
0.12
0.06
0.00
−0.06
−0.12
−0.18
−0.24

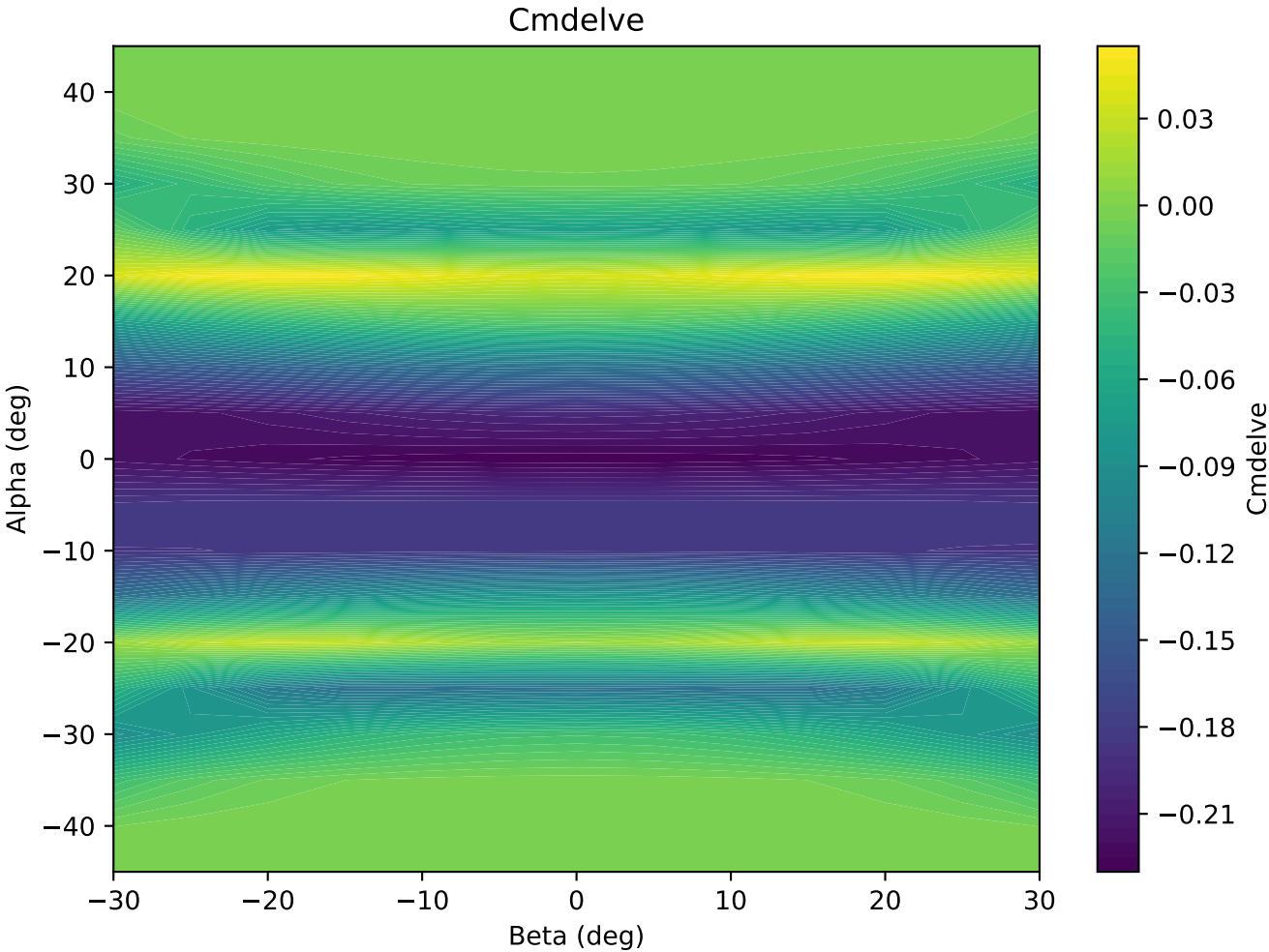

Cldelve
Alpha (deg)
Beta (deg)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.168
0.126
0.084
0.042
0.000
−0.042
−0.084
−0.126
−0.168
Cldelve

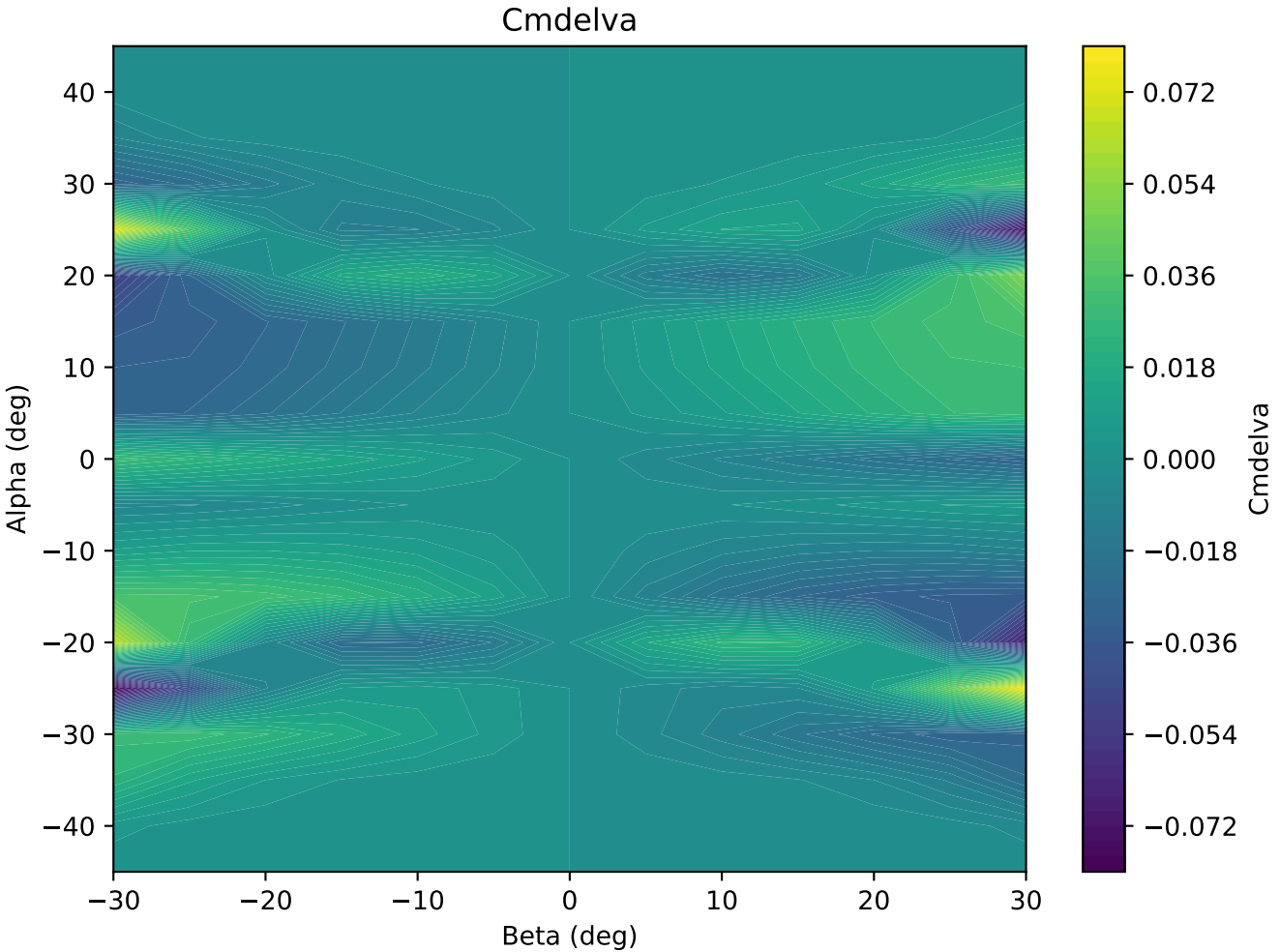

Cldelva
Alpha (deg)
Beta (deg)
Cldelva
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.18
0.12
0.06
0.00
−0.06
−0.12
−0.18
−0.24
−0.30

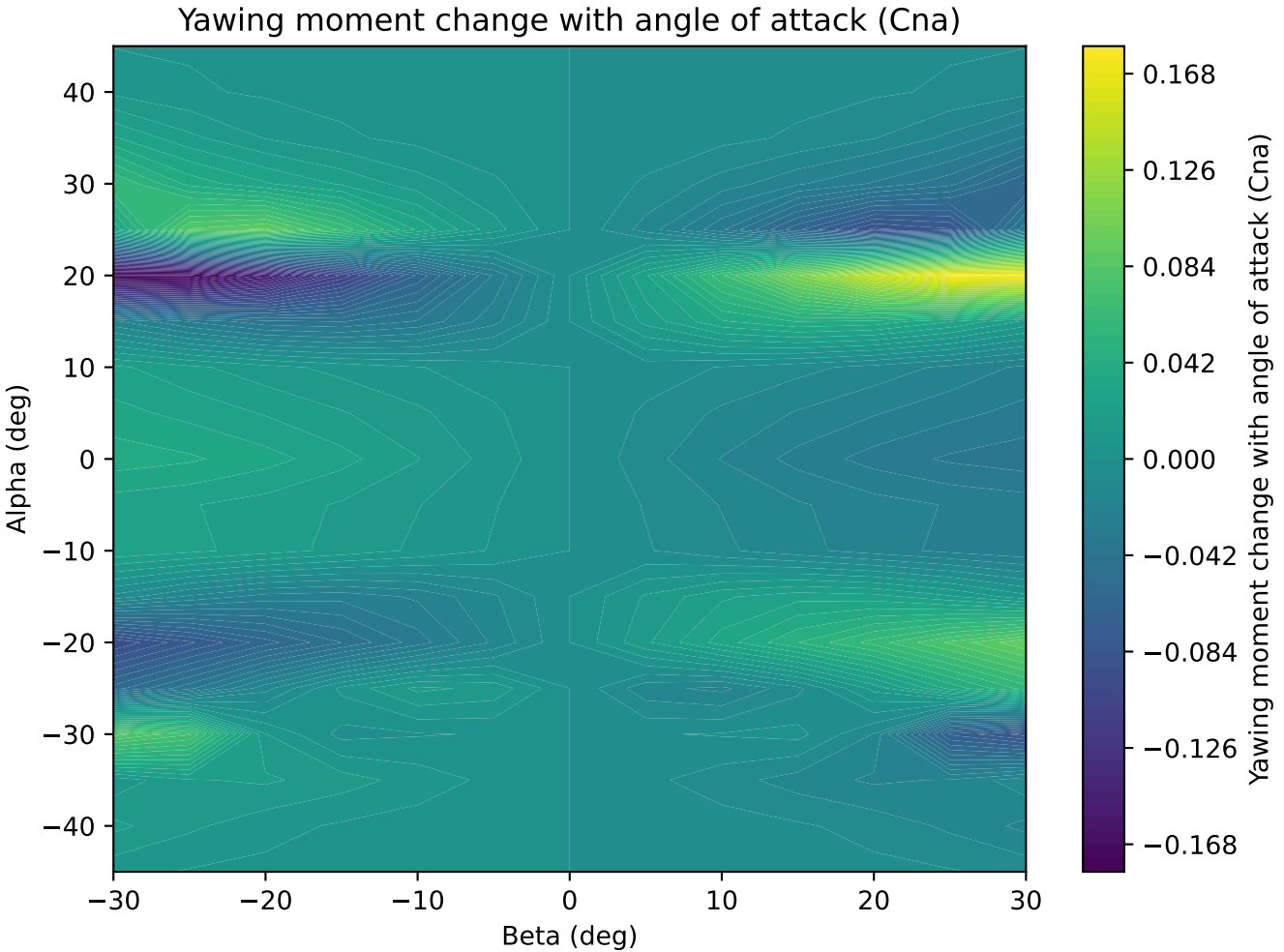

Pitching moment change with sideslip (Cmb)
Alpha (deg)
Beta (deg)
Pitching moment change with sideslip (Cmb)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4

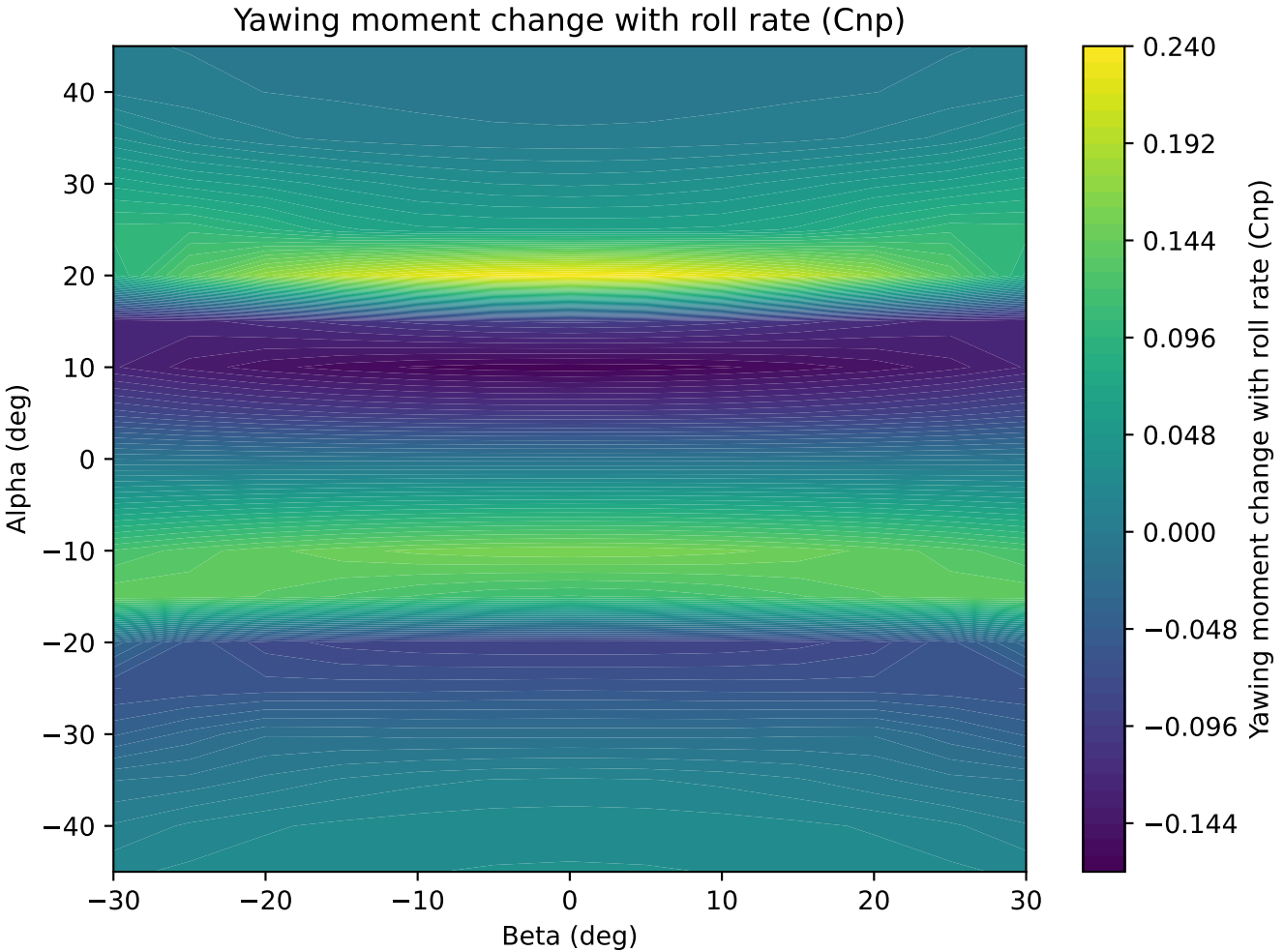

Pitching moment change with roll rate (Cmp)
Alpha (deg)
Beta (deg)
Pitching moment change with roll rate (Cmp)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.192
0.144
0.096
0.048
0.000
−0.048
−0.096
−0.144
−0.192

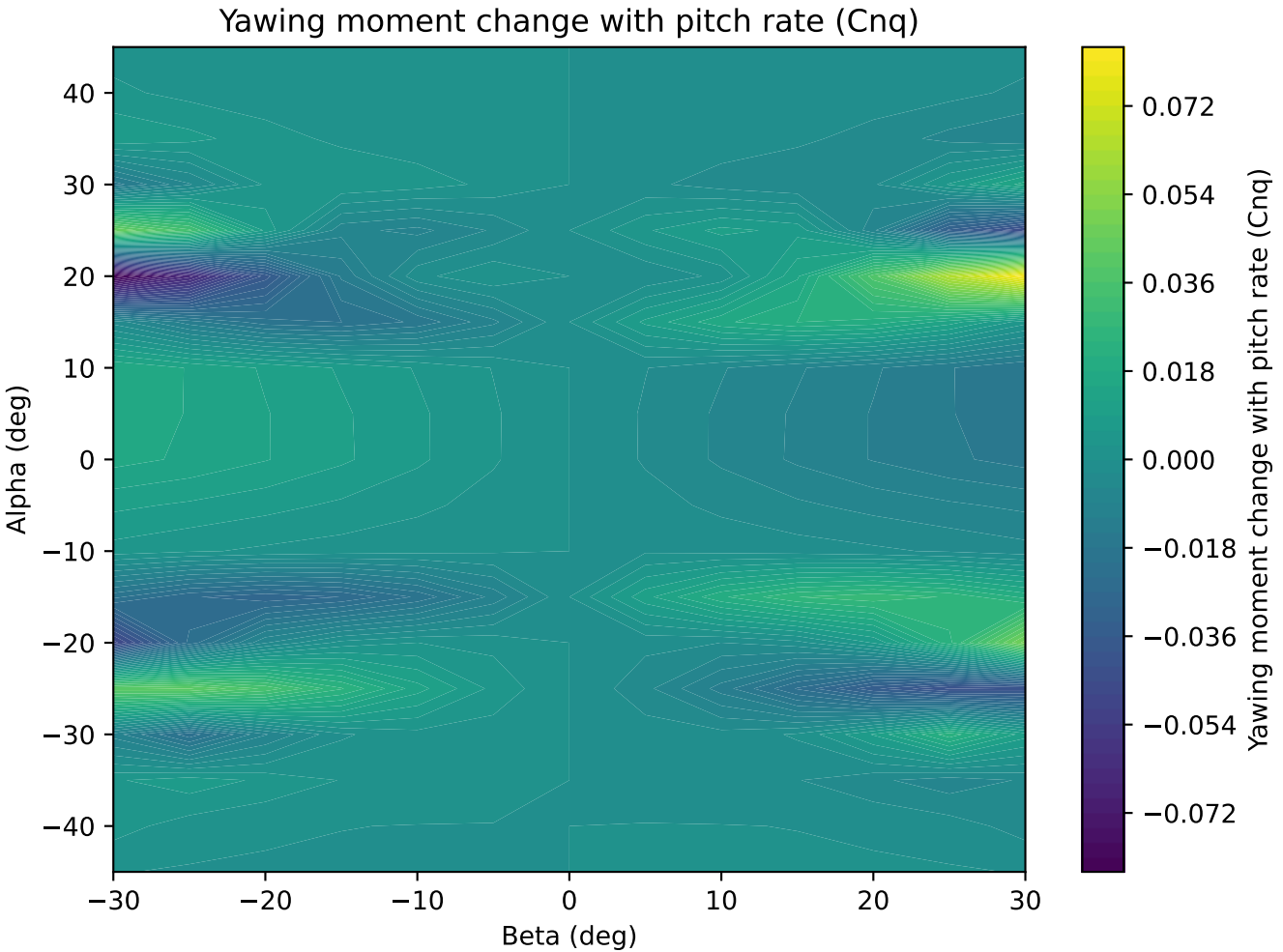

Pitching moment change with yaw rate (Cmr)
Alpha (deg)
Beta (deg)
Pitching moment change with yaw rate (Cmr)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.060
0.045
0.030
0.015
0.000
−0.015
−0.030
−0.045
−0.060

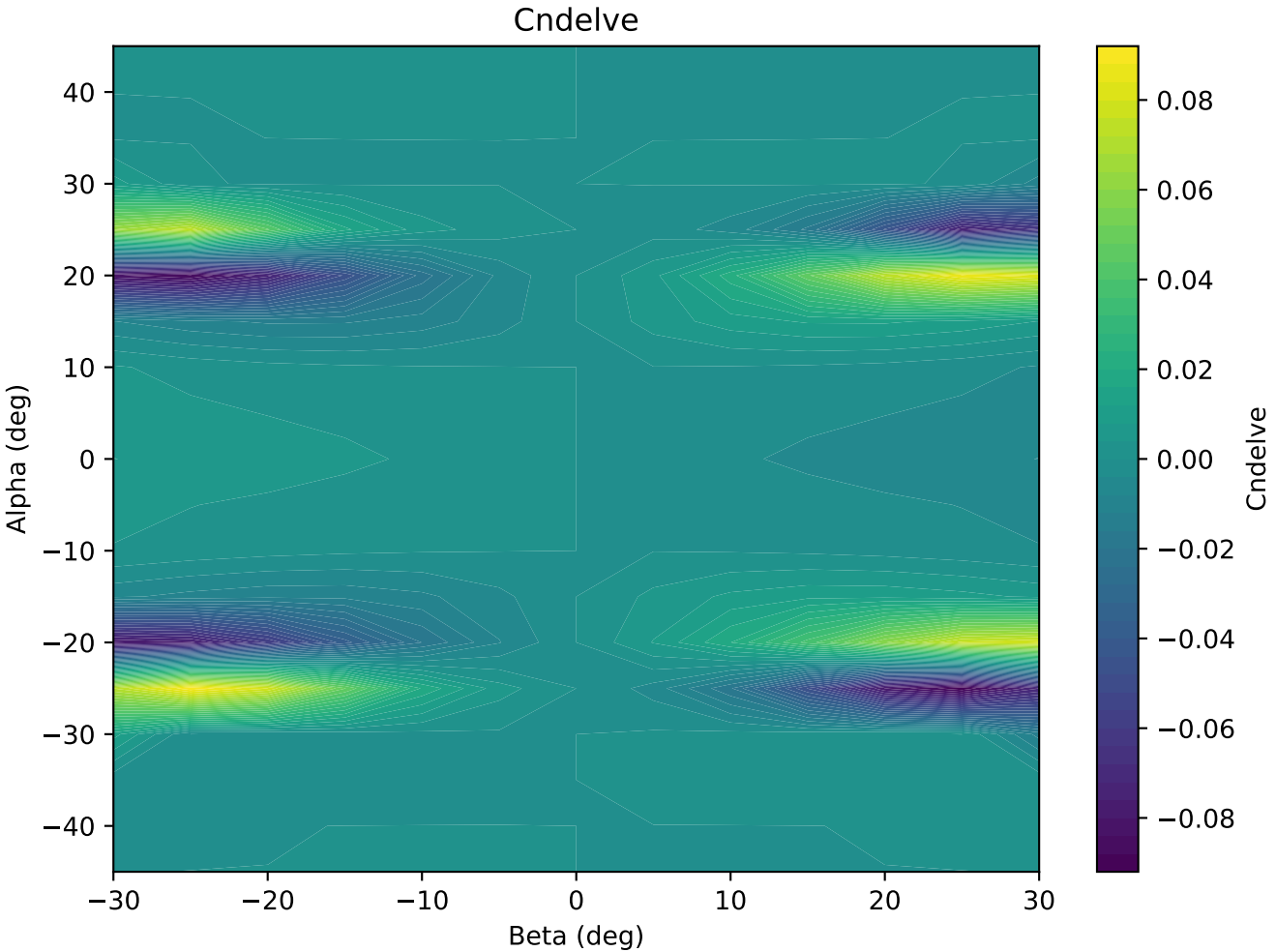

Cmdelve
Alpha (deg)
Beta (deg)
Cmdelve
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.03
0.00
−0.03
−0.06
−0.09
−0.12
−0.15
−0.18
−0.21

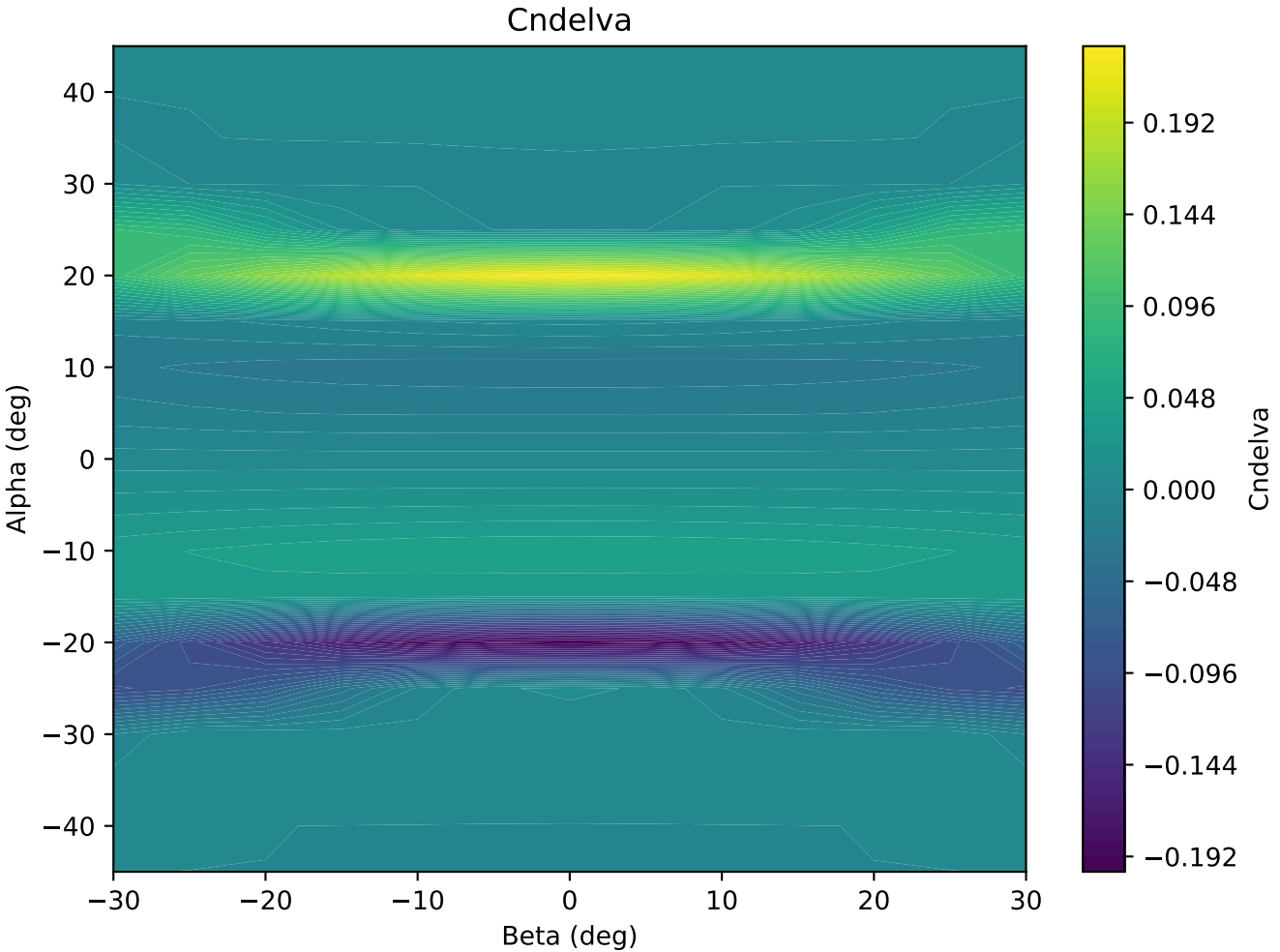

Cmdelva
Alpha (deg)
Beta (deg)
Cmdelva
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.072
0.054
0.036
0.018
0.000
−0.018
−0.036
−0.054
−0.072


Yawing moment change with angle of attack (Cna)
Alpha (deg)
Beta (deg)
Yawing moment change with angle of attack (Cna)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.168
0.126
0.084
0.042
0.000
−0.042
−0.084
−0.126
−0.168


Yawing moment change with roll rate (Cnp)
Alpha (deg)
Beta (deg)
Yawing moment change with roll rate (Cnp)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.240
0.192
0.144
0.096
0.048
0.000
−0.048
−0.096
−0.144


Yawing moment change with pitch rate (Cnq)
Alpha (deg)
Beta (deg)
Yawing moment change with pitch rate (Cnq)
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.072
0.054
0.036
0.018
0.000
−0.018
−0.036
−0.054
−0.072


Cndelve
Alpha (deg)
Beta (deg)
Cndelve
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08


Cndelva
Alpha (deg)
Beta (deg)
Cndelva
40
30
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20
30
0.192
0.144
0.096
0.048
0.000
−0.048
−0.096
−0.144
−0.192